

Learning the Portfolio Objective:

Machine Learning for Implementable U.S. Equity Portfolios

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2026

Declaration

The work contained in this thesis is my own work unless otherwise stated.

Acknowledgements

I would like to thank my supervisor, Professor Johannes Muhle-Karbe, for suggesting the initial idea for this project and for his guidance throughout its development. His advice and our discussions helped me broaden the way I approached the research questions and think more carefully about the economic interpretation of the results.

I am also grateful to the Department of Mathematics and to Imperial College London. Having spent three years here as an undergraduate and a further year as a Master's student, all within the Department of Mathematics, I am very thankful for the education, opportunities, and academic environment I have experienced over these four years.

I would especially like to thank my parents for always supporting me in pursuing the subjects and directions that genuinely interest me. Finally, I am grateful to all the friends who have supported me throughout these years, both within and beyond university, and who have made this period of my life more enjoyable and meaningful.

Abstract

This dissertation examines whether machine learning creates greater implementable value when it is trained directly on the portfolio objective rather than on stock-level return prediction. Using a dynamically reconstituted Top500 U.S. equity universe, the analysis compares prediction-first methods with Linear and Random-Features Portfolio-ML under common risk, transaction-cost, and portfolio-accounting assumptions over 2005–2024. The strongest prediction-first benchmark, Static-ML*, delivers annualised utility of 0.46%, compared with 2.10% for Linear Portfolio-ML and 2.36% for Random Features. The corresponding utility improvements are 1.64 and 1.90 percentage points and are statistically supported in the baseline specification. Random Features adds only 0.26 percentage points of utility relative to Linear, indicating that most of the gain is already captured by the linear Portfolio-ML specification.

The performance advantage arises primarily from higher gross portfolio returns rather than lower realised trading costs. Portfolio-ML also adjusts gradually from inherited holdings, keeps realised risk closer to the common ex-ante scale, and reduces trading intensity as investor size increases. The Portfolio-ML ranking persists across richer information sets and the tested AUM range. At the feature level, momentum has the largest economic importance in both specifications despite relatively low raw persistence, while simple persistence measures do not consistently explain feature-level economic value. Overall, the Portfolio-ML architecture materially improves implementable out-of-sample utility, while additional nonlinear flexibility plays a secondary role.

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Introduction

Machine learning has become an important tool for empirical asset pricing because firm characteristics contain weak, nonlinear, and interacting information about future stock returns that is difficult to capture with a small number of manually specified predictors. Much of this literature formulates the statistical problem at the level of individual securities: characteristics observed at time t are used to estimate expected returns over the next period, and the resulting forecasts are then translated into portfolio positions. This approach is natural when the objective is prediction, but an investor’s economic problem is broader. Portfolio performance depends on how forecasts interact with covariance risk, the investor’s existing positions, the turnover required to exploit a signal, and the cost of establishing and unwinding positions. Once trading costs are material, current holdings become part of the state of the investment problem and the economic value of a signal depends on the portfolio path generated by using it. A model can therefore contain useful predictive information while producing weak implementable performance if the resulting positions require excessive trading or are poorly scaled to risk. This distinction between statistical prediction and portfolio-level economic value is the central motivation for this dissertation.

The empirical methodology builds on the Portfolio-ML framework of [11], which applies machine learning directly to dynamic portfolio choice. Firm characteristics are mapped into an aim portfolio, and the parameters of this mapping are estimated using a portfolio-level objective that incorporates expected portfolio returns, covariance risk, trading costs, and inherited positions. The dynamic structure is closely related to [9], where costly trading leads investors to adjust gradually from current holdings towards a forward-looking aim portfolio. This partial-adjustment mechanism also makes the persistence of investment opportunities economically relevant: establishing a position is more valuable when the associated expected-return opportunity is expected to remain available for longer.

The empirical implementation adapts this framework to an independently constructed U.S. equity setting. Portfolio-ML is compared with a sequence of prediction-first benchmarks that use the same information set and the same risk, transaction-cost, and portfolio-accounting assumptions. These include a return-ranking portfolio, a transaction-cost-aware Static-ML strategy, and a strengthened Static-ML* specification that allows the portfolio stage to adjust the scale of predicted returns, covariance risk, and perceived trading costs. Portfolio-ML is estimated with both a linear characteristic map and a Random-Features specification. This comparison addresses two separate questions: whether direct portfolio-level learning improves implementable out-of-sample performance, and whether additional nonlinear flexibility adds further economic value within Portfolio-ML. Beyond the baseline comparison, the dissertation extends the empirical analysis in three directions. It examines where the Portfolio-ML performance advantage appears in implemented portfolios, tests how the fitted rules adapt to investor scale and broader information sets, and links feature-level economic importance to raw characteristic persistence and predictive-alpha decay. These exercises complement [11] by moving from the baseline performance comparison to the implementation, scalability, and signal-level interpretation of the fitted Portfolio-ML policies.

The empirical analysis uses a dynamically reconstituted universe of 500 large U.S. equities constructed from the JKP Global Factor database and CRSP. The research panel runs from January 1990 through December 2024, portfolio-level estimation begins in 1995, and formal out-of-sample evaluation covers January 2005 through December 2024, providing 240 monthly test observations. The baseline information set contains ten ranked firm characteristics spanning momentum, short-term reversal, risk, size, liquidity, value, profitability, and investment. All formal portfolio methods use the same one-month-ahead excess-return convention, factor-based covariance estimates, stock-specific quadratic trading-cost model based on recent dollar volume, and dynamic treatment of inherited holdings and universe entry and exit. Baseline assets under management are fixed at USD 1 billion and the risk-aversion coefficient is $\gamma = 10$. Model estimation and hyperparameter selection follow a cumulative out-of-sample protocol: each formal test year uses only information and validation outcomes available before that year, while the portfolio state continues across calendar-year boundaries. The comparison therefore focuses on how different learning and portfolio-construction rules translate the same information into positions under the same implementation assumptions.

The baseline results show a large gap between return prediction and implementable portfolio performance. Return-ML Rank earns a small positive annualised gross return of 0.16%, but its average monthly turnover of 46.32% generates annualised trading costs of 1.28%, leaving a net return of -1.12% and annualised utility of -3.55% . Portfolio optimisation improves this outcome substantially. Static-ML* reduces average monthly turnover to 11.23%, earns an annualised net return of 0.87%, and delivers annualised utility of 0.46%. Portfolio-ML improves further. Linear Portfolio-ML earns an annualised net return of 3.14%, with annualised volatility of 4.23%, a net Sharpe ratio of 0.743, and annualised utility of 2.10%. The Random-Features specification earns a net return of 3.86%, with volatility of 5.21%, a net Sharpe ratio of 0.742, and utility of 2.36%. The main ranking is therefore clear: transaction-cost-aware portfolio construction materially improves the economic use of return forecasts, and the Portfolio-ML architecture improves further on the strongest prediction-first implementation.

The utility differences are also supported by formal out-of-sample inference. Linear Portfolio-ML exceeds Static-ML* by 1.64 percentage points of annualised utility, with a Newey–West statistic of 2.09 and a circular block-bootstrap 95% confidence interval of $[0.20\%, 3.06\%]$. Random-Features Portfolio-ML exceeds Static-ML* by 1.90 percentage points, with a HAC statistic of 2.02 and a bootstrap interval of $[0.09\%, 3.63\%]$. Both Portfolio-ML specifications deliver higher annual utility than Static-ML* in 15 of the 20 formal test years. Most of the Portfolio-ML gain is already captured by the linear specification. Random Features raises annualised utility by a further 0.26 percentage points relative to Linear, with a HAC statistic of 0.42 and a bootstrap interval of $[-1.01\%, 1.37\%]$. The additional nonlinear flexibility raises the return point estimate but contributes little incremental utility relative to the main improvement from direct portfolio-level learning. The broader comparison supports the economic relevance of the Portfolio-ML design, which combines portfolio-level estimation with dynamic aim-based adjustment. The empirical comparison identifies the value of this combined Portfolio-ML design rather than separately attributing the gain to portfolio-level estimation and dynamic adjustment.

Most of the Portfolio-ML net-return advantage is already present before realised trading costs are deducted. Relative to Static-ML*, Linear Portfolio-ML earns 2.22 percentage points more annualised gross return but saves only 0.05 percentage points of annualised trading cost, so realised cost savings account for approximately 2% of the net-return difference. For Random Features, the corresponding cost-saving share is less than 1%. The dominant source of the observed performance gap is the higher gross return generated by the Portfolio-ML rules rather than lower realised trading costs. The implementation

diagnostics show that this advantage is accompanied by a materially different portfolio path: Portfolio-ML trades gradually from inherited positions towards a forward-looking aim, uses materially lower trading intensity in the Linear specification, and keeps realised portfolio risk closer to the common ex-ante scale. The average implemented monthly trade covers 28.5% of the immediate L1 distance from inherited holdings to the Linear aim portfolio and 26.4% for Random Features; in transaction-cost-weighted distance, the corresponding values are 13.7% and 16.9%. Full-sample realised-to-predicted volatility ratios are 0.93 for Linear and 0.95 for Random Features, compared with 1.37 for Static-ML*. Together, these results show that Portfolio-ML converts the same underlying information into a materially different sequence of positions, combining stronger gross portfolio returns with gradual adjustment and more stable portfolio scaling.

The AUM experiment gives a further view of how the portfolio rules respond when market impact becomes more important. The analysis extends the cumulative portfolio experiment across fixed AUM levels from zero to USD 10 billion. Linear Portfolio-ML turnover falls from 18.75% per month in the zero-cost diagnostic to 4.74% at USD 1 billion and 2.74% at USD 10 billion; Random-Features turnover falls from 69.28% to 11.42% and then 5.36%. Both specifications retain positive net returns and positive utility throughout the tested AUM range, and Linear becomes marginally more attractive than Random Features at the USD 10 billion endpoint. The relative advantage over Static-ML* narrows rather than expands with scale: the Linear utility difference falls from 3.23 percentage points in the zero-cost diagnostic to 1.36 percentage points at USD 10 billion. As trading costs rise, both Portfolio-ML specifications respond by trading less and reducing portfolio scale, while preserving the baseline ranking over the AUM range considered.

The main ranking also survives the two richer information sets considered in the robustness analysis. Under Final15 and Final20 Conservative, the Linear-minus-Static* utility differences are 1.53 and 0.95 percentage points, while the corresponding Random-Features differences are 1.89 and 1.25 percentage points. Final20 Conservative is constructed using pre-formal-OOS information from 1995–2004 and does not use 2005–2024 portfolio outcomes in feature selection. Portfolio-ML remains ahead of Static-ML* outside the Core10 baseline, although the estimates are less precise than under Core10. The magnitude also varies materially across years, with 2007 making the largest positive contribution. In the available Final15 evaluation-only exclusion, removing the realised 2007 observations reduces the Linear utility advantage to approximately zero and the Random-Features advantage to 0.51 percentage points. The broader result is a persistent ranking whose magnitude varies across information sets and periods.

The final extension examines which characteristics matter economically inside the fitted Portfolio-ML policies. A fixed-model economic feature-importance diagnostic neutralises one ranked characteristic at a time and recursively recomputes the portfolio path while holding fitted model parameters and economic inputs fixed. Momentum has the largest positive economic feature importance in both Portfolio-ML specifications, followed by historical volatility, while several slow-moving characteristics have weak or negative fixed-model importance. The analysis then compares this economic dependence with two notions of persistence: persistence of the raw characteristic state and persistence of its univariate relation with future returns. Raw characteristic persistence has only a weak cross-feature relation with economic importance, with Spearman correlations of 0.21 for Linear and -0.09 for Random Features. A multi-horizon predictive-alpha measure produces a stronger positive relation for Random Features, with a correlation of 0.43, but only 0.12 for Linear and no stable relation across the alternative prespecified metrics. Momentum is the clearest example: it ranks only ninth among the ten Core10 characteristics in raw state persistence, yet it has the highest economic importance in both Portfolio-ML specifications and retains positive predictive content over several future horizons. The

results therefore do not reveal a stable one-dimensional relationship between signal persistence and economic importance. The feature-level results instead suggest that economic value depends on how predictive content interacts with covariance risk, trading costs, inherited positions, and the remaining signals.

This dissertation takes the Portfolio-ML framework of [11] as its methodological starting point and makes three main empirical contributions. First, it provides an independent implementation and out-of-sample evaluation of Portfolio-ML in a dynamic large-cap U.S. equity setting, using progressively stronger prediction-first benchmarks under common return, risk, cost, and portfolio-accounting conventions. This design isolates the comparison between prediction-first and portfolio-level learning while keeping the underlying information and implementation environment common. Second, the analysis examines where the Portfolio-ML performance advantage appears in implemented portfolios. The gross-return decomposition, trading-intensity and liquidity diagnostics, inherited-position adjustment, risk calibration, and AUM sensitivity experiments show how the fitted portfolio rules translate the same characteristics into different realised portfolio paths. Third, the dissertation adds a signal-level interpretation that links fixed-model economic feature importance to both raw characteristic persistence and predictive-alpha decay. This extension separates three objects that are often treated together—state persistence, predictive persistence, and portfolio-level economic value—and shows that their empirical rankings differ materially. These additions complement [11] by moving from the baseline Portfolio-ML comparison to a more detailed empirical account of implementation, scalability, robustness, and feature-level economic interpretation.

The remainder of the dissertation develops this analysis in detail. Chapter 1 reviews the literature on machine-learning return prediction, mean–variance portfolio choice, transaction costs, dynamic trading, Portfolio-ML, and signal persistence. Chapter 2 describes the data, dynamic Top500 universe, firm characteristics, transaction-cost inputs, and factor risk model. Chapter 3 presents the five formal portfolio methods and the cumulative out-of-sample estimation, selection, accounting, and inference procedures. Chapter 4 reports the baseline out-of-sample performance and the risk-aversion sensitivity analysis. Chapter 5 examines the economic mechanisms and the response of the portfolio rules to investor scale. Chapter 6 studies feature-set and temporal robustness, economic feature importance, characteristic persistence, and predictive-alpha decay. The final chapter summarises the empirical findings and their implications for the use of machine learning in implementable portfolio choice.

Chapter 1

Literature Review and Economic Framework

1.1 Machine Learning for Return Prediction

A large empirical literature studies whether firm characteristics contain information about the cross-section of future stock returns. Machine learning extends this literature by allowing expected returns to depend on many characteristics through flexible and potentially nonlinear functions. This is particularly useful in equity markets, where predictive information may be distributed across numerous weak signals and interactions that are difficult to specify *ex ante*. Empirical studies such as [10], [8], and [2] document the ability of flexible statistical methods to extract return-predictive structure from high-dimensional firm characteristics.

Let $s_{i,t} \in \mathbb{R}^K$ denote the vector of characteristics observed for stock i at time t , and let $r_{i,t+1}$ denote its excess return over the following period. A conventional prediction model estimates a function f such that

$$f(s_{i,t}) \approx \mathbb{E}_t[r_{i,t+1}].$$

Under squared-error loss, a generic estimator solves

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{|\mathcal{I}|} \sum_{(i,t) \in \mathcal{I}} (r_{i,t+1} - f(s_{i,t}))^2,$$

possibly subject to regularisation. At the population level, the target is the conditional mean. Linear models impose a restricted mapping from characteristics to returns, while nonlinear methods allow richer interactions and conditional relationships; in both cases the statistical objective remains future return prediction.

The fitted forecasts form the cross-sectional vector

$$\hat{\mu}_t = \left(\hat{f}(s_{1,t}), \dots, \hat{f}(s_{N,t}) \right)',$$

which is then translated into portfolio positions. A simple implementation ranks securities by predicted return, while a more structured approach combines $\hat{\mu}_t$ with a covariance matrix in a portfolio optimisation problem. This prediction-first sequence—estimate returns first, construct the portfolio second—is the natural benchmark considered by [11].

The economic objective of the investor is broader than the forecasting loss. Portfolio value depends on covariance risk, existing positions, the turnover required to exploit a signal, and the cost of changing positions. Two forecasts with similar statistical accuracy can therefore have very different investment value. This point becomes especially important once trading costs are introduced. Empirical work such as [4] and [5] shows that

transaction costs can materially change both the profitability of characteristic strategies and the relative attractiveness of competing models.

The framework of [11] moves this economic objective into the learning problem itself. Instead of rewarding a model for stock-level forecast accuracy and imposing portfolio considerations only afterwards, Portfolio-ML estimates the characteristic-to-aim-portfolio mapping under an objective defined in terms of portfolio return, risk, trading costs, and inherited holdings. The empirical analysis in this dissertation therefore compares prediction-first methods with Portfolio-ML under a common implementation environment, separating the common Portfolio-ML architecture from the value of additional model flexibility.

1.2 Portfolio Choice and Mean–Variance Optimisation

The link from expected-return forecasts to portfolio holdings begins with mean–variance portfolio choice [13]. Let

$$\mu_t = \mathbb{E}_t[r_{t+1}]$$

denote the vector of conditional expected excess returns and

$$\Sigma_t = \text{Var}_t(r_{t+1})$$

the conditional covariance matrix. For risky-asset weights π_t , a mean–variance investor with risk-aversion parameter $\gamma > 0$ solves

$$\max_{\pi_t} \left\{ \pi_t' \mu_t - \frac{\gamma}{2} \pi_t' \Sigma_t \pi_t \right\}.$$

With unconstrained risky-asset positions and a risk-free asset absorbing the residual wealth, the optimal risky portfolio is

$$\pi_t^M = \frac{1}{\gamma} \Sigma_t^{-1} \mu_t,$$

the frictionless Markowitz portfolio. Expected returns determine the direction of desired exposures, while the covariance matrix determines their scale after accounting for diversification and common risk. A large expected return need not imply a large position if the asset adds substantial portfolio variance, while several individually weak forecasts can be valuable when their risks diversify well.

Varying risk aversion traces the conventional efficient frontier. In the unconstrained frictionless case,

$$\mathbb{E}_t[r_{t+1}^\pi] = \frac{1}{\gamma} \mu_t' \Sigma_t^{-1} \mu_t,$$

and portfolio volatility is

$$\sigma_t^\pi = \frac{1}{\gamma} \sqrt{\mu_t' \Sigma_t^{-1} \mu_t}.$$

The Sharpe ratio,

$$\frac{\mathbb{E}_t[r_{t+1}^\pi]}{\sigma_t^\pi} = \sqrt{\mu_t' \Sigma_t^{-1} \mu_t},$$

does not depend on γ . Risk aversion scales exposure to the same underlying risky portfolio, which gives the familiar straight-line investment opportunity set when a risk-free asset is available.

This frictionless benchmark also highlights what changes once implementation matters. With costless rebalancing, the optimal position at time t depends on current expected returns and covariance, not on the portfolio inherited from the previous period. A prediction-first model fits naturally into this setup: machine learning estimates $\hat{\mu}_t$, and a separate optimiser combines it with Σ_t . Trading costs make that separation incomplete because the current position and the path of future adjustment become part of the decision.

1.3 Transaction Costs and Dynamic Portfolio Choice

Trading costs turn portfolio choice into an intertemporal problem. If changing a position is costly, the value of a desired portfolio depends on the trade required to reach it as well as on expected return and risk. Early work studies portfolio selection with proportional or fixed transaction costs [3]; later work links return predictability to rebalancing costs [1]. A particularly tractable formulation is provided by [9], who study predictable returns in a multi-asset setting with quadratic trading costs.

Let Δx_t denote the vector of trades at time t . A quadratic transaction-cost function takes the form

$$TC_t(\Delta x_t) = \frac{1}{2} \Delta x_t' \Lambda_t \Delta x_t,$$

where Λ_t is positive semidefinite. Under a diagonal specification, the cost of each trade depends only on its own size. Quadratic costs capture market impact by making aggressive rebalancing increasingly expensive.

A schematic dynamic objective is

$$\mathbb{E}_0 \left[\sum_{t \geq 0} \delta^t \left(x_t' \mu_t - \frac{\gamma}{2} x_t' \Sigma x_t - \frac{1}{2} (x_t - x_{t-1})' \Lambda (x_t - x_{t-1}) \right) \right].$$

The position chosen today now affects both current utility and future rebalancing costs. A large move towards a high-return opportunity may be unattractive if the opportunity is short-lived or likely to reverse before the trade can earn back its implementation cost.

Gârleanu and Pedersen show that the optimal policy has a partial-adjustment form. Rather than moving immediately to the current frictionless Markowitz portfolio, the investor trades toward an *aim portfolio*:

$$x_t = x_{t-1} + M_{\text{rate}} (\text{aim}_t - x_{t-1}),$$

where M_{rate} governs the speed of adjustment. Existing holdings therefore retain value when the benefit of immediate rebalancing is smaller than its cost.

The aim portfolio is itself forward-looking. When trading costs are proportional to risk, Gârleanu and Pedersen express it as a weighted average of the current and expected future Markowitz portfolios,

$$\text{aim}_t = \sum_{h=0}^{\infty} z(1-z)^h \mathbb{E}_t [\text{Markowitz}_{t+h}], \quad 0 < z < 1.$$

The target incorporates where investment opportunities are expected to move, not only the frictionless portfolio that would be optimal today. This is their “aim in front of the target” principle. It also gives persistence a direct economic role: an opportunity that survives for longer can receive greater weight in the forward-looking aim portfolio because its expected benefit persists over more future periods.

For empirical portfolios, the inherited state must also account for asset returns. If securities held over the previous period earn different returns, the investor begins the next period with a grown portfolio rather than the weights originally chosen. Following [11], let $g_t\pi_{t-1}$ denote this grown inherited portfolio. If w_t is assets under management, the dollar trade required to reach π_t is

$$\tau_t = w_t (\pi_t - g_t\pi_{t-1}).$$

With quadratic market impact, transaction costs as a fraction of wealth are

$$TC_t^\pi = \frac{w_t}{2} (\pi_t - g_t\pi_{t-1})' \Lambda_t (\pi_t - g_t\pi_{t-1}).$$

This formulation makes two implementation features explicit. Trades are measured relative to the grown inherited portfolio, and the same change in portfolio weights becomes more expensive as investor size increases. Current holdings, expected future opportunities, and the cost of changing exposures therefore enter the portfolio decision jointly.

1.4 Portfolio-ML and the Implementable Efficient Frontier

The dynamic problem creates a difficult state space for conventional machine learning. With trading costs, the optimal position at time t depends on current characteristics, inherited holdings, and the expected evolution of future opportunities. A fully general policy could be written as

$$\pi_t = \pi(\dots, s_{t-2}, s_{t-1}, s_t),$$

but direct estimation quickly becomes impractical. [11] use the structure of the dynamic portfolio problem to reduce this dimensionality.

Their starting point is a first-order approximation to the optimal dynamic policy:

$$\pi_t = mg_t\pi_{t-1} + (I - m)A_t + O(\epsilon^2),$$

where A_t is the aim portfolio and m depends on risk, trading costs, and portfolio growth. The relevant transformation of m has eigenvalues between zero and one, so the portfolio retains part of the grown inherited position and adjusts only gradually toward the aim. This extends the partial-adjustment principle in [9].

The aim portfolio is forward-looking. In the general formulation of [11],

$$A_t = (I - m)^{-1} \sum_{h=0}^{\infty} (m\Lambda^{-1}\bar{g}\Lambda)^h c \mathbb{E}_t \left[\frac{1}{\gamma} \Sigma^{-1} \mu(s_{t+h}) \right],$$

for a matrix c determined by the model primitives. The expression inside the expectation is the frictionless Markowitz portfolio associated with future investment opportunities. As in Gârleanu and Pedersen, the economically relevant aim depends on the expected path of future opportunities and the cost of adjusting toward them.

One implementation is to estimate expected returns separately at several future horizons and combine them according to the dynamic solution. Jensen et al. call this Multiperiod-ML. The empirical analysis here uses their alternative Portfolio-ML formulation, which learns the mapping from current characteristics directly to the aim portfolio. Suppose

$$A_t = f(s_t)\beta,$$

where $f(s_t)$ transforms the current characteristics and β is a parameter vector. A linear specification uses the original characteristics, while a nonlinear specification can use a richer basis. The nonlinear implementation considered here follows [11] and uses random features in the sense of [16].

Substituting the aim rule into the dynamic recursion gives

$$\pi_t = \tilde{s}_t \beta,$$

where $\tilde{s}_t$ collects current and lagged transformed characteristics after accounting for portfolio growth and partial adjustment. The dynamic optimisation can then be written as a quadratic problem in β . In the notation of Jensen et al., its sample analogue is

$$\mathcal{U}(\beta) = E_T[\tilde{r}_{t+1}]' \beta - \frac{1}{2} \beta' E_T[\tilde{\Sigma}_t] \beta,$$

where $E_T[\tilde{r}_{t+1}]$ captures the return contribution of the transformed portfolio exposures and $E_T[\tilde{\Sigma}_t]$ combines risk and trading-cost penalties. With ridge regularisation,

$$\hat{\beta} = \left(E_T[\tilde{\Sigma}_t] + \lambda I \right)^{-1} E_T[\tilde{r}_{t+1}].$$

The key difference from prediction-first learning lies in this estimation objective. The coefficient vector is selected on portfolio returns, risk, and trading costs rather than on stock-level prediction error. A static transaction-cost-aware method can still use a strong portfolio stage—estimating $\hat{\mu}_t$ first and then choosing π_t with covariance risk, inherited holdings, and trading costs in the objective—but the return model remains separate from the economic consequences of the positions it generates. This comparison motivates the strengthened Static-ML* benchmark used later in the dissertation.

Transaction costs also change the relevant efficient frontier. With market impact, scaling a portfolio raises trading costs disproportionately, so expected net return and volatility no longer move proportionally with risk aversion. Jensen et al. define the *implementable efficient frontier* for target risk σ as

$$k(\sigma) = \max_{\pi \in \Pi} \mathbb{E} \left[r_{t+1}^{\pi, \text{net}} \right]$$

subject to

$$\mathbb{E} [\pi_t' \Sigma \pi_t] = \sigma^2.$$

The frontier records the best expected return net of trading costs at each risk level. It also depends on investor size: larger dollar trades face greater market impact, and the optimal policy can respond by trading less, reducing scale, or shifting exposures. The empirical analysis therefore evaluates the competing methods across a range of risk-aversion coefficients and AUM levels, testing the portfolio rules under different degrees of implementation pressure.

1.5 Signal Persistence, Alpha Decay, and Economic Feature Importance

Transaction costs make the time dimension of predictability relevant for portfolio value. A signal that disappears quickly leaves less time for the investor to recover the cost of establishing and unwinding a position, while a persistent return opportunity can support a larger aim position. This is the role of alpha decay in [9]: slowly mean-reverting return-predicting states influence expected future Markowitz portfolios for longer and receive greater weight in the aim portfolio.

The persistence of the observed characteristic is not the same object as the persistence of the return opportunity. Let $s_{i,t}^{(j)}$ denote characteristic j for stock i . A characteristic may move slowly over time while its association with future returns decays quickly. Conversely, a faster-moving characteristic may contain information about several future return horizons. The economically relevant predictive object is the horizon profile

$$\mathbb{E}_t[r_{i,t+h}] \text{ as a function of } s_{i,t}^{(j)}, \quad h = 1, 2, \dots$$

Portfolio-ML introduces a third object: the value of a characteristic inside the fitted portfolio policy. Because characteristics enter a multivariate portfolio problem, their contribution can depend on return predictability, predictive horizon, interactions with other signals, covariance exposure, liquidity, inherited holdings, and the trading required to express the signal. Raw state persistence, predictive persistence, and portfolio-level economic value need not rank characteristics in the same order.

Jensen et al. formalise feature value through *economic feature importance*. Let $V(s)$ denote maximum attainable utility with the full information set s , and let s^{-j} remove feature j . The population importance of feature j is

$$\iota_j = V(s) - V(s^{-j}).$$

For small transaction costs, Jensen et al. provide the approximation

$$\iota_j = [v(s) - v(s^{-j})] - [c(s) - c(s^{-j})] + O(\|\Lambda\|^2),$$

where $v(s)$ is frictionless portfolio value and $c(s)$ captures the trading cost associated with variation in the frictionless target. A feature can contribute through the investment opportunity it identifies, through the cost of adjusting positions to exploit it, or through both channels. The persistence result in Gârleanu and Pedersen applies to return-predicting states, so the theory points toward predictive persistence rather than mechanically toward high autocorrelation of the raw characteristic.

The empirical extension in Chapter 6 keeps these objects separate. Economic feature importance is measured by neutralising one characteristic while holding the fitted Portfolio-ML policy and economic inputs fixed, then recomputing the recursive portfolio path. Characteristic persistence measures how slowly the observed rank state evolves, while predictive-alpha decay tracks the univariate relation between a current characteristic and returns at future horizons. The resulting comparison asks whether simple measures of persistence help explain which characteristics matter economically inside the fitted portfolio policy.

Chapter 2

Data and Empirical Setting

This chapter describes the empirical environment used throughout the dissertation. The analysis combines stock returns and trading data from CRSP with firm characteristics from the JKP Global Factor database developed by [12]. The baseline investment universe consists of a dynamically reconstituted set of 500 large U.S. equities. Portfolio decisions are made at a monthly frequency, using characteristics observed at month-end and returns over the following month. The main portfolio analysis spans 1995–2024, while formal out-of-sample evaluation covers 2005–2024.

2.1 Data Sources and Investment Universe

The starting point is the U.S. sample of the JKP Global Factor database. The database provides firm characteristics, market equity, and one-month-ahead excess returns. These data are merged with CRSP monthly and daily security data using `permno`. The eligible cross-section consists of U.S. observations satisfying the JKP common-security, main-exchange, primary-security, and main-observation indicators. CRSP provides monthly prices and security identifiers, together with the daily return and trading-volume data required for liquidity measurement and covariance estimation.

The investable universe is reconstructed at each month-end. For stock i in month t , the month-end price is defined as the absolute CRSP price,

$$P_{i,t} = |\text{prc}_{i,t}|,$$

and securities are required to satisfy

$$P_{i,t} \geq 5.$$

Among eligible securities passing this screen, stocks are ranked in descending order by the JKP market-equity field $ME_{i,t}$, and the 500 largest securities are retained. If $\text{rank}_{i,t}^{ME}$ denotes the descending market-capitalisation rank within the eligible monthly cross-section, the investment-universe indicator can be written as

$$I_{i,t}^{500} = \mathbf{1} \{P_{i,t} \geq 5, \text{rank}_{i,t}^{ME} \leq 500\}.$$

Membership is recomputed independently each month, so the identity of the securities in the portfolio universe changes through time. The resulting panel contains exactly 500 securities in each month from January 1990 to December 2024. The full research-ready panel therefore contains 420 monthly cross-sections and 210,000 stock-month observations. The realised sample includes securities traded on the NYSE, AMEX, and Nasdaq.

The dynamic Top500 construction concentrates the analysis on relatively large and liquid U.S. equities while retaining changes in the investment universe through time. The first

five years of the panel provide historical information for return-model estimation, liquidity measurement, and risk-model initialisation. Portfolio-level analysis begins in January 1995. Formal out-of-sample evaluation begins in January 2005 and ends in December 2024, giving 240 monthly cross-sections and 120,000 stock-month observations.

2.2 Firm Characteristics

The baseline specification uses ten firm characteristics, referred to as Core10. The information set is deliberately compact while spanning several economically distinct dimensions of the cross-section: momentum, short-term reversal, risk, size and liquidity, value, profitability, and investment.

Most characteristics are taken directly from the JKP database. Market capitalisation and trading liquidity are also used to construct local size and liquidity variables. Table 2.1 summarises the baseline information set.

Table 2.1: Baseline Core10 characteristics

Characteristic	Source field	Theme	Interpretation
<code>momentum_12.1</code>	JKP <code>ret_12.1</code>	Momentum	12-to-1 past return
<code>reversal_1m</code>	JKP <code>ret_1.0</code>	Reversal	Most recent one-month return
<code>volatility_12m</code>	JKP <code>rvol_252d</code>	Risk / Low volatility	Historical return volatility
<code>beta_mkt</code>	JKP <code>beta_60m</code>	Risk / Low volatility	Market beta
<code>idio_vol</code>	JKP <code>ivol_capm_60m</code>	Risk / Low volatility	CAPM idiosyncratic volatility
<code>log_mcap</code>	$\log(ME)$	Liquidity / Size	Log market capitalisation
<code>adv_6m</code>	CRSP daily data	Liquidity / Size	Six-month average dollar volume
<code>book_to_market</code>	JKP <code>be_me</code>	Value	Book-to-market ratio
<code>gross_profitability</code>	JKP <code>gp_at</code>	Quality / Profitability	Gross profitability relative to assets
<code>asset_growth</code>	JKP <code>at_gr1</code>	Investment / Accruals	Asset growth

The accounting-based characteristics follow the construction and timing conventions embedded in the JKP database, whose underlying accounting information is based on Compustat. The resulting variables enter the empirical panel in the same form as the other JKP characteristics.

The liquidity variable plays two distinct roles. Raw six-month average dollar volume enters the transaction-cost model described below. A cross-sectionally ranked version of the same quantity enters Core10 as a predictive characteristic. Portfolio-learning models therefore observe relative liquidity as part of the information set, whereas trading costs depend on the underlying dollar trading volume.

Richer characteristic sets are considered separately in the robustness analysis. Core10 remains the baseline information set for the principal model comparisons and economic mechanism analysis.

2.3 Return Timing and Characteristic Preprocessing

The predictive target is the JKP-supplied one-month-ahead excess stock return,

$$r_{i,t+1}^e.$$

In the processed panel, this variable is stored as `ret_exc_lead1m`. The field attached to the month-end t observation represents the excess return earned over the following month. The information timing throughout the analysis is therefore

$$s_{i,t} \longrightarrow r_{i,t+1}^e,$$

where $s_{i,t}$ denotes the vector of characteristics observed on the month- t row.

The same forward excess-return field is used as the target in the return-prediction models and as the risky-asset return in formal portfolio evaluation. This gives the prediction-first and portfolio-level methods a common return convention.

Before entering the models, raw characteristics are transformed into monthly cross-sectional ranks. These ranks are formed within the broader JKP-eligible cross-section before the sample is restricted to the largest 500 securities. The transformed variables preserve a Top500 stock's relative position within the wider eligible U.S. equity cross-section rather than only its position among the final 500 names.

For characteristic j , let $x_{i,t}^{(j)}$ denote its raw value and let $N_t^{(j)}$ denote the number of non-missing observations in the eligible cross-section. For an observed value, the normalised characteristic is

$$s_{i,t}^{(j)} = \frac{\text{rank}_t(x_{i,t}^{(j)}) - 1}{N_t^{(j)} - 1}.$$

Ranks are ascending and tied observations receive their average rank. The transformation therefore maps valid observations into the interval $[0, 1]$. Missing characteristics are assigned the neutral value

$$s_{i,t}^{(j)} = 0.5.$$

The stored rank-transformed characteristics form the common feature representation used by the empirical models. Model-specific transformations, including the standardisation used by the linear return-prediction model and the construction of the Portfolio-ML design matrix, are introduced with the corresponding estimation procedures.

This timing convention separates the observed information set from the subsequent return realisation. At each month-end, the investment universe and characteristic vector are determined from the current data row, and the associated portfolio is evaluated using the corresponding one-month-ahead excess return.

2.4 Transaction-Cost Inputs

Implementation costs are modelled using a quadratic market-impact specification. Following the calibration used by [11], which is motivated by the institutional trading-cost evidence in [7], the transaction-cost matrix is diagonal and depends inversely on recent dollar trading volume.

For stock i on trading day d , daily dollar volume is calculated from CRSP as

$$DV_{i,d} = |\text{prc}_{i,d}| |\text{vol}_{i,d}|.$$

Daily dollar volume is first averaged within each calendar month. Let $\overline{DV}_{i,t}$ denote this monthly average. Six-month average dollar volume is then defined as

$$ADV_{i,t}^{6m} = \frac{1}{K_{i,t}} \sum_{k=0}^{K_{i,t}-1} DV_{i,t-k}, \quad K_{i,t} = \min\{6, \text{available monthly observations}\}.$$

In the mature sample, we have $K_{i,t} = 6$. The current month is included in the calculation because portfolio formation takes place at month-end.

The stock-specific quadratic-impact coefficient is

$$\Lambda_{i,t} = \frac{0.2}{ADV_{i,t}^{6m}},$$

and the transaction-cost matrix is

$$\Lambda_t = \text{diag}(\Lambda_{1,t}, \dots, \Lambda_{N,t}).$$

For dollar assets under management W and a vector of changes in portfolio weights $\Delta\pi_t$, transaction costs expressed as a fraction of wealth take the form

$$TC_t = \frac{W}{2} \Delta\pi_t' \Lambda_t \Delta\pi_t.$$

This specification assigns greater market impact to a given dollar trade in securities with lower average dollar volume, while more liquid stocks can absorb larger trades at lower cost. The quadratic form also makes implementation costs increase with investor scale.

The baseline empirical specification fixes assets under management at

$$W = \$1 \text{ billion.}$$

This dollar scale enters both transaction-cost-aware portfolio construction and realised cost accounting. Portfolio holdings evolve through time as security returns change the inherited positions, while the scalar baseline AUM used to scale market impact remains fixed. Alternative capital scales are considered in the AUM sensitivity analysis.

2.5 Risk Model

All transaction-cost-aware portfolio methods require an estimate of the conditional covariance matrix of stock returns. Estimating an unrestricted 500×500 covariance matrix directly from monthly returns would be noisy and poorly conditioned. The analysis instead uses a factor-based covariance model in the spirit of the empirical risk specification in [11].

For month t , let B_t denote the matrix of observable stock-level factor exposures. The conditional covariance matrix is represented as

$$\Sigma_t = B_t F_t B_t' + D_t,$$

where F_t is the covariance matrix of factor returns and D_t is a diagonal matrix of idiosyncratic variances.

The exposure matrix contains an intercept, one-digit SIC industry indicators, and three style exposures:

$$\text{size,} \quad \text{12-to-1 momentum,} \quad \text{return volatility.}$$

The corresponding characteristic fields are

$$\text{rank_log_mcap,} \quad \text{rank_momentum_12_1,} \quad \text{rank_volatility_12m.}$$

Within each month, each style exposure is winsorised at the 1st and 99th cross-sectional percentiles and standardised to zero mean and unit population standard deviation. The industry variables are constructed from one-digit SIC classifications, with one industry category omitted as the reference group.

Factor returns are estimated from daily CRSP stock returns. When a delisting return is available, it is incorporated into the corresponding daily return. Daily returns are then clipped to

$$[-0.5, 0.5]$$

before the cross-sectional factor regression is estimated.

For trading day d , stock returns are related to factor exposures measured at the preceding month-end:

$$r_d = B_d f_d + \varepsilon_d.$$

The daily factor-return estimate is obtained from the ridge-regularised cross-sectional regression

$$\hat{f}_d = \left(\frac{1}{N_d} B_d' B_d + \alpha P \right)^{-1} \frac{1}{N_d} B_d' r_d,$$

where

$$\alpha = 0.001,$$

and the intercept is excluded from the ridge penalty through the matrix P .

The factor covariance matrix is updated daily using an exponentially weighted moving-average recursion with span 60. Defining

$$a = \frac{2}{60 + 1},$$

the factor-return mean is updated according to

$$\mu_d = \mu_{d-1} + a \left(\hat{f}_d - \mu_{d-1} \right).$$

Let

$$\delta_d = \hat{f}_d - \mu_{d-1}.$$

The factor covariance recursion is

$$F_d = (1 - a) \left[F_{d-1} + a \delta_d \delta_d' \right].$$

The corresponding decay rate implies a half-life of approximately 20.8 trading days.

Idiosyncratic variances are estimated from the squared daily residuals using the same EWMA span. If $q_{i,d}$ denotes the idiosyncratic variance state for stock i ,

$$q_{i,d} = (1 - a) q_{i,d-1} + a \varepsilon_{i,d}^2.$$

At each month-end, the latest daily factor covariance and idiosyncratic-variance states are converted to monthly units using 21 trading days. The stock-level covariance matrix supplied to the portfolio models is therefore

$$\Sigma_t = 21 \left(B_t F_t B_t' + \text{diag}(q_t) \right).$$

The assembled covariance matrix is symmetrised and its eigenvalues are bounded below by 10^{-6} , providing a positive-definite covariance input for portfolio optimisation.

The factor-risk inputs cover all 360 portfolio-analysis months from January 1995 through December 2024.

2.6 Empirical Sample Summary

Table 2.2 summarises the baseline empirical setting.

All subsequent portfolio methods are evaluated in this common empirical environment: they face the same dynamic investment universe, characteristic information, forward-return target, stock-specific implementation costs, and conditional risk estimates. The comparison can therefore focus on how each method translates the same information, risk, and transaction-cost inputs into portfolio positions.

Table 2.2: Baseline empirical setting

Item	Baseline specification
Asset class	U.S. equities
Portfolio frequency	Monthly
Investment universe	Dynamic monthly Top500
Minimum price	USD 5
Master panel	1990–2024
Master-panel observations	210,000 stock-months
Main portfolio-analysis period	1995–2024
Main-analysis observations	180,000 stock-months
Formal out-of-sample period	2005–2024
Formal OOS observations	120,000 stock-months
Baseline information set	Core10
Predictive target	One-month-ahead excess return
Characteristic transformation	Monthly cross-sectional rank in $[0, 1]$
Missing-characteristic value	0.5
Trading-cost input	Six-month raw dollar ADV
Baseline AUM	USD 1 billion
Risk model	Factor covariance plus idiosyncratic variance

Chapter 3

Methodology

This chapter describes how the common empirical inputs developed in the previous chapter are translated into portfolio positions. The formal comparison contains five methods: Return-ML Rank, Static-ML, Static-ML*, Portfolio-ML Linear, and Portfolio-ML Random Features. The first three methods use a common return-prediction layer, whereas Portfolio-ML estimates the mapping from characteristics to an aim portfolio directly and combines that aim with inherited holdings through the dynamic adjustment rule. All methods are evaluated under the same return, risk, transaction-cost, and portfolio-accounting conventions.

3.1 Empirical Portfolio Objective and Accounting

Let π_t denote the vector of portfolio weights selected at month-end t , and let r_{t+1} denote the vector of one-month-ahead excess stock returns associated with that decision date. The gross portfolio excess return is

$$R_{t+1}^{\text{gross}} = \pi_t' r_{t+1}.$$

Because portfolio positions are carried through time, trading is measured relative to the portfolio inherited at the beginning of the month rather than relative to zero. Let h_t denote the vector of grown inherited weights for securities that remain in the current investment universe, and define the current-universe trade as

$$\Delta_t = \pi_t - h_t.$$

Securities that leave the dynamic Top500 universe are liquidated when they exit. If ℓ_t denotes the corresponding liquidation trade, the realised transaction cost is

$$TC_t = \frac{W}{2} (\Delta_t' \Lambda_t \Delta_t + \ell_t' \Lambda_{t-1} \ell_t),$$

where W is assets under management and Λ_t is the diagonal quadratic-impact matrix defined previously. The second term is zero in months without exits. The baseline analysis uses

$$W = \$1 \text{ billion.}$$

Net portfolio return is therefore

$$R_{t+1}^{\text{net}} = R_{t+1}^{\text{gross}} - TC_t.$$

Portfolio risk is evaluated using the conditional covariance matrix Σ_t available at the portfolio date. For risk-aversion coefficient γ , the monthly utility flow used throughout the formal evaluation is

$$U_{t+1} = R_{t+1}^{\text{net}} - \frac{\gamma}{2} \pi_t' \Sigma_t \pi_t,$$

or equivalently,

$$U_{t+1} = R_{t+1}^{\text{gross}} - TC_t - \frac{\gamma}{2} \pi_t' \Sigma_t \pi_t.$$

The baseline specification sets

$$\gamma = 10.$$

This criterion follows the economic structure of [11]: realised portfolio returns are evaluated net of implementation costs, while portfolio risk is measured using the ex-ante covariance matrix that enters the investment decision. The main utility statistic reported in the empirical analysis is the annualised mean monthly utility flow,

$$\bar{U}^{\text{ann}} = 12 \frac{1}{T} \sum_{t=1}^T U_{t+1}.$$

The same accounting convention is applied to every formal method, so differences in performance come from the portfolio rule used to determine π_t , not from method-specific definitions of returns, risk, or transaction costs. The prediction-first methods begin by estimating one-month-ahead expected returns, which motivates the common return-prediction layer described next.

3.2 Return-Prediction Layer

Return-ML provides the expected-return estimates used by Return-ML Rank, Static-ML, and Static-ML*. Let

$$x_{i,t} = \left(s_{i,t}^{(1)}, \dots, s_{i,t}^{(K)} \right)'$$

denote the $K = 10$ ranked Core10 characteristics for stock i at month-end t . The prediction target is the JKP one-month-ahead excess return

$$r_{i,t+1}^e.$$

The return-prediction problem is therefore

$$x_{i,t} \longrightarrow \hat{\mu}_{i,t} \approx \mathbb{E}_t \left[r_{i,t+1}^e \right].$$

Two model classes are considered: linear Ridge regression and a nonlinear Random-Features Ridge specification. Hyperparameters are selected according to out-of-sample prediction mean squared error rather than portfolio performance. The resulting annual prediction stream is then used unchanged by each of the prediction-first portfolio methods.

3.2.1 Linear Ridge Prediction

For the linear specification, the ranked characteristics are standardised using the mean and population standard deviation estimated from the training sample. Let X denote the resulting standardised design matrix after an intercept column has been added, and let y denote the vector of one-month-ahead excess returns. The coefficient estimator is

$$\hat{\theta}_\lambda = \left(\frac{1}{n} X'X + \lambda P \right)^+ \frac{1}{n} X'y,$$

where

$$P = \text{diag}(0, 1, \dots, 1),$$

so that the intercept is not penalised. The superscript $+$ denotes the Moore–Penrose inverse used in the implementation.

The corresponding prediction for stock i is

$$\hat{\mu}_{i,t} = \tilde{x}'_{i,t} \hat{\theta}_{\lambda},$$

where $\tilde{x}_{i,t}$ contains the intercept and the training-sample-standardised characteristics.

The candidate Ridge penalties are

$$\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}.$$

3.2.2 Random-Features Ridge Prediction

The second specification introduces nonlinear dependence on the characteristics through Random Features in the sense of [16]. For an even number p of final Random Features, $p/2$ projection vectors are drawn as

$$w_j \sim N(0, \eta^2 I), \quad j = 1, \dots, p/2.$$

Let X denote the rank-normalised characteristic matrix used by the return-prediction model, with missing or non-finite characteristic values replaced by 0.5. Unlike the linear Ridge specification, these characteristic columns are not standardised using training-sample means and standard deviations before the Random-Features transformation. If W denotes the matrix containing the projection vectors, the feature map is

$$\Phi_p(X) = \frac{1}{\sqrt{p}} [\cos(XW), \sin(XW)].$$

Thus p denotes the total number of sinusoidal feature columns.

An intercept is added to the transformed design. No additional centering or scaling is applied to the Random-Features columns. The coefficients are then estimated using the same Ridge objective as above, with the intercept left unpenalised. The Random-Features candidates use

$$p \in \{32, 64\}, \quad \eta \in \{0.1, 0.5\},$$

together with the same Ridge grid

$$\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}.$$

The random-number seed is fixed at 20260710, so the feature maps are reproducible across estimation years and candidate Ridge penalties.

3.2.3 Annual Prediction Selection

Return-prediction hyperparameters are selected separately for each calendar prediction year. For year Y , the available estimation history ends in November of year $Y - 1$. The most recent 120 months are used as the desired validation window, provided that sufficient earlier data remain for model estimation. At least 24 months are retained for initial training and at least 12 months are required for validation.

Each candidate model is fitted on observations preceding the validation window and evaluated by mean squared prediction error,

$$MSE = \frac{1}{N_{\text{val}}} \sum_{(i,t) \in \mathcal{V}} (r_{i,t+1}^e - \hat{\mu}_{i,t})^2.$$

The candidate with the lowest validation MSE is selected. The chosen specification is then refitted using all prediction data available through November $Y - 1$, and the resulting model produces predictions for each month of year Y .

Because the return-history panel begins in January 1990, all formal test years from 2005 onward use the full ten-year prediction-validation window. The annual prediction procedure is entirely upstream of portfolio construction: the same selected predictions are supplied to Return-ML Rank, Static-ML, and Static-ML*. Portfolio-level hyperparameter selection is treated separately below.

3.3 Return-ML Rank

Return-ML Rank provides the simplest mapping from predicted returns to portfolio positions. It isolates the economic value of the return-prediction layer before introducing covariance or transaction-cost-aware portfolio optimisation.

At each month-end, the predicted returns

$$\hat{\mu}_{1,t}, \dots, \hat{\mu}_{N_t,t}$$

are ordered cross-sectionally. Let $q_{i,t}$ denote the resulting zero-based ordinal rank of stock i , with larger ranks corresponding to larger predicted returns. Define

$$\bar{q}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} q_{i,t}.$$

The portfolio weight is

$$\pi_{i,t}^{\text{Rank}} = L \frac{q_{i,t} - \bar{q}_t}{\sum_{j=1}^{N_t} |q_{j,t} - \bar{q}_t|},$$

where the gross-exposure parameter is fixed at

$$L = 1.$$

For a non-degenerate cross-section, this construction implies

$$\sum_{i=1}^{N_t} \pi_{i,t}^{\text{Rank}} = 0$$

and

$$\sum_{i=1}^{N_t} \left| \pi_{i,t}^{\text{Rank}} \right| = 1.$$

The portfolio therefore has zero net investment and unit gross exposure. Stocks with relatively high predicted returns receive positive weights, while those with relatively low predicted returns receive negative weights.

The target weights at month-end t are determined entirely by the current cross-sectional return ranking. Once these weights have been formed, realised trading costs are calculated using the common dynamic accounting convention: current positions are compared with the grown portfolio inherited from the previous month, and securities leaving the Top500 universe are liquidated. Risk and transaction costs enter the evaluation of the strategy even though the target portfolio itself is formed directly from predicted returns.

This benchmark is useful because it separates return forecasting from portfolio implementation. It asks whether a model that successfully orders stocks by expected return also produces economically useful positions when the resulting trades are evaluated under the same risk-and-cost environment as the more sophisticated portfolio methods.

3.4 Static-ML

Return-ML Rank uses the predicted-return ordering directly and therefore provides a useful benchmark for the predictive content of the characteristics. Static-ML introduces portfolio optimisation as a second stage. Expected returns are first estimated by the common Return-ML procedure, after which the portfolio is chosen by balancing predicted return, ex-ante risk, and the cost of trading away from the inherited position.

Let $\hat{\mu}_t$ denote the vector of one-month-ahead expected returns produced by Return-ML and let h_t denote the grown inherited portfolio aligned with the current investment universe. For numerical stability, define

$$\Sigma_t^{\text{reg}} = \Sigma_t + 10^{-8}I.$$

Static-ML solves the one-period problem

$$\max_{\pi_t} \left\{ s\hat{\mu}_t'\pi_t - \frac{\gamma}{2}\pi_t'\Sigma_t^{\text{reg}}\pi_t - \frac{W}{2\phi}(\pi_t - h_t)'\Lambda_t(\pi_t - h_t) \right\},$$

where s controls the scale of the predicted-return vector and ϕ controls the strength of the trading-cost penalty.

The first-order condition gives the closed-form portfolio

$$\pi_t^{\text{Static}} = \left[\gamma\Sigma_t^{\text{reg}} + \frac{W}{\phi}\Lambda_t \right]^{-1} \left[s\hat{\mu}_t + \frac{W}{\phi}\Lambda_th_t \right].$$

The rule has a simple interpretation. The expected-return term pulls the portfolio towards positions favoured by the current forecasts, whereas the transaction-cost term pulls it towards the inherited portfolio. A larger value of ϕ weakens the latter force and permits more rapid adjustment towards the current target.

The baseline implementation considers

$$\phi \in \{1, 3, 6, 12\}$$

and

$$s \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1\}.$$

These portfolio parameters are selected from their cumulative out-of-sample histories using the procedure described below. The underlying return-prediction specification remains determined separately by prediction MSE.

Static-ML is transaction-cost aware but remains prediction-first. Characteristics first enter a statistical model of expected returns; the resulting $\hat{\mu}_t$ is then treated as an input to a separate one-period portfolio problem. The inherited portfolio affects the current decision through the trading-cost term, but the optimisation does not incorporate the forward-looking dynamic adjustment matrix used by Portfolio-ML.

3.5 Static-ML*

Portfolio performance in a prediction-first strategy can depend strongly on the scale of estimated returns, risks, and trading costs. Following the additional portfolio-tuning approach of [11], the formal comparison therefore includes a more flexible static benchmark, denoted Static-ML*.

Starting from the same Return-ML expected-return vector, Static-ML* constructs the decision inputs

$$\hat{\mu}_t^* = u\hat{\mu}_t,$$

$$\Sigma_t^* = \Sigma_t + v \operatorname{diag}(\operatorname{diag}(\Sigma_t)),$$

and

$$\Lambda_t^* = k\Lambda_t.$$

Here u adjusts the scale of expected returns, v increases the diagonal component of the decision covariance matrix, and k rescales the transaction-cost matrix used in portfolio construction.

The resulting portfolio is

$$\pi_t^{\text{Static}^*} = [\gamma(\Sigma_t^* + 10^{-8}I) + Wk\Lambda_t]^{-1} [u\hat{\mu}_t + Wk\Lambda_t h_t].$$

The candidate parameter values are

$$u \in \{0.05, 0.1, 0.25, 0.5, 1\},$$

$$v \in \{0, 0.25, 0.5, 1\},$$

and

$$k \in \{0.5, 1, 2\}.$$

The triple (u, v, k) is chosen from cumulative out-of-sample portfolio histories. These adjustments determine the portfolio decision. Once positions have been formed, returns, transaction costs, and ex-ante risk are evaluated using the common baseline accounting defined above, with the original Λ_t and repaired Σ_t .

Static-ML* provides the most flexible prediction-first benchmark in the formal comparison. It allows the second-stage portfolio problem to adjust the scale of expected returns, covariance risk, and perceived trading costs without altering the return-prediction model itself. Despite this additional flexibility, expected returns are still estimated first and portfolio weights second; Portfolio-ML instead learns the portfolio mapping directly from the economic objective.

3.6 Portfolio-ML

Portfolio-ML replaces the two-stage prediction-and-optimisation architecture with direct learning of the aim-portfolio mapping. Following [11], the method parameterises an aim portfolio as a function of the current characteristics and estimates that function by maximising a portfolio-level objective that includes risk and transaction costs.

Let S_t denote the transformed characteristic design matrix at month-end t . The aim portfolio is parameterised as

$$A_t = S_t \beta,$$

where β is a vector of coefficients to be estimated. Given the grown inherited portfolio h_t , the corresponding actual portfolio is

$$\pi_t = m_t h_t + (I - m_t) A_t,$$

where m_t is the full adjustment matrix implied by the dynamic trading-cost problem. The matrix determines how rapidly each component of the inherited portfolio is adjusted towards the current aim.

For estimation, the inherited portfolio can itself be represented as a linear function of β . Let H_t denote the grown inherited loading matrix. Define

$$B_t = m_t H_t + (I - m_t) S_t.$$

The portfolio can then be written as

$$\pi_t = B_t \beta.$$

This representation is the empirical counterpart of the transformed signal representation developed in the theoretical framework: current and past portfolio decisions are absorbed into a loading matrix, leaving a quadratic objective in the common coefficient vector β .

Let

$$\Delta B_t = B_t - H_t$$

denote the loading change associated with current-universe trading. If securities leave the investment universe, let E_t collect their grown inherited loading rows. The transformed return vector is

$$\tilde{r}_{t+1} = B_t' r_{t+1},$$

and the transformed quadratic penalty is

$$\tilde{\Sigma}_t = \gamma B_t' \Sigma_t B_t + W(\Delta B_t)' \Lambda_t \Delta B_t + W E_t' \Lambda_{t-1} E_t,$$

where the final term applies when securities exit the current universe.

For a given feature specification and ridge parameter λ , Portfolio-ML estimates β by maximising

$$Q_\lambda(\beta) = \beta' \left(\frac{1}{T} \sum_{t=1}^T \tilde{r}_{t+1} \right) - \frac{1}{2} \beta' \left[\frac{1}{T} \sum_{t=1}^T \tilde{\Sigma}_t + \lambda I \right] \beta.$$

The resulting estimator has the closed form

$$\hat{\beta}_\lambda = \left[\frac{1}{T} \sum_{t=1}^T \tilde{\Sigma}_t + \lambda I \right]^{-1} \left(\frac{1}{T} \sum_{t=1}^T \tilde{r}_{t+1} \right).$$

The key difference from the prediction-first methods is visible directly in this objective. The coefficients are chosen according to the economic contribution of the resulting portfolio positions rather than according to stock-level prediction errors. Return information, covariance risk, current trading costs, and the cost of liquidating exiting positions all enter the same estimation problem. The coefficient ridge regularises the resulting portfolio map and is applied to the full coefficient vector.

3.6.1 Linear Portfolio-ML

The linear specification uses the Core10 characteristic ranks directly. Let X_t denote the $N_t \times K$ matrix of ranked characteristics. Within each month, every characteristic is first demeaned cross-sectionally and then divided by its cross-sectional Euclidean norm. If $C_t(X_t)$ denotes this transformation and

$$c_t = \frac{1}{\sqrt{N_t}} \mathbf{1}_{N_t},$$

a unit-norm constant column is added to the design.

The complete design is subsequently scaled by current stock-level volatility derived from the same covariance model used in portfolio construction. Writing

$$\sigma_t = \sqrt{\text{diag}(\Sigma_t)},$$

the linear design can be represented as

$$S_t^{\text{Linear}} = \text{diag}(\sigma_t)^{-1} [c_t, C_t(X_t)].$$

This transformation gives each characteristic a comparable cross-sectional scale and reduces the mechanical influence of stocks with high conditional volatility. The constant column is treated as an ordinary portfolio characteristic and is included in the coefficient ridge.

For Linear Portfolio-ML, the ridge parameter is selected from

$$\lambda \in \{0.1, 1, 10, 100, 1000, 10000\}.$$

3.6.2 Random-Features Portfolio-ML

The nonlinear specification first maps the Core10 ranks into a Random-Features representation. For $p/2$ Gaussian projection vectors,

$$w_j \sim N(0, \eta^2 I), \quad j = 1, \dots, p/2,$$

the initial nonlinear design is

$$Z_t = \frac{1}{\sqrt{p}} [\cos(X_t W), \sin(X_t W)].$$

As in the return-prediction model, p denotes the total number of sinusoidal columns.

After the Random-Features transformation, each column is demeaned cross-sectionally and normalised by its monthly Euclidean norm. The same $1/\sqrt{N_t}$ constant column is then added, and the full design is scaled by current stock-level volatility. Hence

$$S_t^{\text{RF}} = \text{diag}(\sigma_t)^{-1} [c_t, C_t(Z_t)].$$

The formal Random-Features specification uses

$$p \in \{32, 64\}, \quad \eta \in \{0.1, 0.5\},$$

and

$$\lambda \in \{0.1, 1, 10, 100, 1000, 10000\}.$$

The random seed is fixed at 20260710, producing a common, reproducible feature bank across estimation years. For a given value of η , the $p = 32$ specification uses the corresponding prefix of the $p = 64$ projection bank.

The Linear and Random-Features specifications differ only in the flexibility of the characteristic map entering the aim portfolio. Both are estimated through the same portfolio-level quadratic criterion, use the same dynamic adjustment mechanism, and are evaluated under the same risk and transaction-cost accounting.

3.7 Dynamic Portfolio Implementation

The portfolio rules above are implemented in a dynamic investment universe. At each month-end, the decision concerns both the target position for the current set of securities and the portfolio inherited from the previous month.

Let $\pi_{i,t-1}$ denote the previous portfolio weight of stock i , and let $r_{i,t}$ denote the return realised between the previous and current portfolio dates. The grown pre-trade weight is

$$h_{i,t} = \pi_{i,t-1} \frac{1 + r_{i,t}}{G_t},$$

where G_t normalises portfolio weights for aggregate market growth. The primary implementation uses a value-weighted market-growth measure,

$$G_t = \max \left\{ 1 + \frac{\sum_i M E_{i,t-1} r_{i,t}}{\sum_i M E_{i,t-1}}, 0.1 \right\}.$$

This growth adjustment changes the inherited portfolio weights but not the scalar AUM used to measure market impact, which remains fixed at USD 1 billion throughout the baseline analysis.

The dynamic Top500 universe requires explicit treatment of changes in membership. A security entering the universe has inherited weight zero. A security remaining in the universe inherits its grown prior position, while a security leaving the universe is liquidated. If I_t denotes the current universe and D_t the set of securities held previously but absent at date t , then

$$\Delta_{i,t} = \pi_{i,t} - h_{i,t}, \quad i \in I_t,$$

and the liquidation trade for an exiting security is

$$\ell_{i,t} = -h_{i,t}, \quad i \in D_t.$$

The corresponding trading cost is

$$TC_t = \frac{W}{2} \left[\sum_{i \in I_t} \Lambda_{i,t} \Delta_{i,t}^2 + \sum_{i \in D_t} \Lambda_{i,t-1} h_{i,t}^2 \right].$$

Thus, portfolio turnover and implementation costs reflect both ordinary rebalancing and changes in the investable universe.

At the beginning of the formal test sequence, the inherited portfolio is initialised using market-capitalisation weights. Thereafter the portfolio state evolves continuously from one month to the next. Annual model updating does not reset the investor's existing positions; a strategy entering January of a new test year therefore inherits the portfolio held at the end of the preceding December.

3.7.1 The Dynamic Adjustment Matrix

Portfolio-ML uses the full adjustment matrix implied by the dynamic trading-cost problem in [11]. This generalises a scalar partial-adjustment rule by allowing the speed of adjustment to depend jointly on the covariance structure and the cross-section of trading costs.

For each month, define

$$\bar{g} = 1 + r^f + \mu^m,$$

where the implementation uses

$$r^f = 0, \quad \mu^m = 0.007$$

at a monthly frequency. Let

$$X_t = W^{-1} \Lambda_t^{-1/2} \gamma \Sigma_t \Lambda_t^{-1/2},$$

and define

$$\Sigma_t^g = \bar{g}^{-2} (\bar{g}^2 \mathbf{1}\mathbf{1}' + \Sigma_t),$$

together with

$$Y_t = \text{diag} (1 + \text{diag}(\Sigma_t^g)).$$

The initial approximation is obtained from

$$\begin{aligned} \hat{\Sigma}_t &= X_t + 2I, \\ \tilde{m}_t^{(0)} &= \frac{1}{2} \left[\hat{\Sigma}_t - \left(\hat{\Sigma}_t^2 - 4I \right)^{1/2} \right]. \end{aligned}$$

The matrix is then updated by the fixed-point recursion

$$\tilde{m}_t^{(j+1)} = \left[X_t + Y_t - \tilde{m}_t^{(j)} \odot \Sigma_t^g \right]^{-1},$$

where $\odot$ denotes elementwise multiplication. Following the empirical implementation of [11], ten fixed-point updates are used. The adjustment matrix is finally recovered as

$$m_t = \Lambda_t^{-1/2} \tilde{m}_t \Lambda_t^{1/2}.$$

The resulting m_t is a full $N_t \times N_t$ matrix and is recomputed from the current covariance and transaction-cost inputs. For the baseline Top500 universe, its dimension is therefore 500×500 each month. This matrix enters both Portfolio-ML estimation and subsequent portfolio implementation through

$$\pi_t = m_t h_t + (I - m_t) A_t.$$

The same ordering of securities is maintained across the covariance matrix, cost matrix, feature design, and inherited portfolio by aligning all objects on the security identifier before the monthly portfolio problem is solved.

3.8 Cumulative Out-of-Sample Design

The empirical design separates model estimation, hyperparameter selection, and formal performance evaluation through time. The prediction history begins in 1990, portfolio estimation begins in 1995, genuinely out-of-sample portfolio validation begins in 1996, and the formal test period runs from 2005 through 2024.

There are two distinct layers of tuning. The first concerns the Return-ML prediction model. As described in Section 3.2, the prediction specification for year Y is selected using only information available before that year, with the estimation sample ending in November of $Y - 1$. The selected model is then refitted on all available prediction data through that date and used to generate the return forecasts for year Y .

The second layer concerns portfolio construction. For every portfolio candidate h and every validation year V from 1996 onward, a genuinely out-of-sample portfolio history is constructed. For the Static methods, the candidate uses the Return-ML prediction stream available for year V . For Portfolio-ML, the coefficient vector for candidate h is estimated using portfolio data from January 1995 through November $V - 1$, after which the candidate is traded out-of-sample over year V . Each candidate retains its own portfolio state through the validation sequence.

Consider a formal test year Y . Only validation observations dated before Y are available for portfolio-hyperparameter selection. The eligible validation years are therefore

$$V = 1996, \dots, Y - 1.$$

A score is calculated from the accumulated out-of-sample history of each candidate, the highest-scoring candidate is selected, and the required model is refitted using information through November $Y - 1$. Formal performance is then recorded from January through December of year Y .

The November cutoff reflects the one-month-ahead return target. A December $Y - 1$ observation contains a forward return realised during the following month and is therefore not part of the information set available for model estimation before year Y . The same timing rule is applied throughout the formal test period.

The portfolio-selection criterion follows the structure of each method. For Static-ML and Static-ML*, let $\mathcal{V}_Y$ denote all candidate validation months before test year Y . For candidate h , define

$$\bar{R}_h^{\text{gross}} = \frac{1}{|\mathcal{V}_Y|} \sum_{t \in \mathcal{V}_Y} R_{h,t}^{\text{gross}},$$

and

$$\overline{TC}_h = \frac{1}{|\mathcal{V}_Y|} \sum_{t \in \mathcal{V}_Y} TC_{h,t}.$$

The cumulative Static score is

$$S_Y^{\text{Static}}(h) = \bar{R}_h^{\text{gross}} - \overline{TC}_h - \frac{\gamma}{2} \left[\frac{1}{|\mathcal{V}_Y|} \sum_{t \in \mathcal{V}_Y} \left(R_{h,t}^{\text{gross}} \right)^2 - \left(\bar{R}_h^{\text{gross}} \right)^2 \right].$$

Thus, the second-stage tuning of the Static methods uses their realised out-of-sample return and cost histories together with the realised gross-return variance.

For Portfolio-ML, candidate selection instead uses the average common monthly utility flow,

$$\mathcal{S}_Y^{\text{PML}}(h) = \frac{1}{|\mathcal{V}_Y|} \sum_{t \in \mathcal{V}_Y} U_{h,t}.$$

The selected candidate is

$$\hat{h}_Y = \arg \max_h \mathcal{S}_Y(h).$$

This cumulative design means that the validation history available for selection expands through time. For example, the 2010 formal portfolio uses portfolio-validation outcomes from 1996 through 2009. Its Return-ML estimation history ends in November 2009, while Portfolio-ML is refitted on portfolio data from January 1995 through November 2009. The resulting strategy is then evaluated over January–December 2010, with the existing December 2009 portfolio carried into the first decision of the new year.

The formal 2005–2024 sample consists of 240 monthly out-of-sample portfolio observations. At every point in this sequence, model estimation and hyperparameter selection precede the return realisation on which formal performance is evaluated.

3.9 Performance Measurement and Statistical Inference

Portfolio performance is evaluated from monthly gross returns, transaction costs, net returns, risk, turnover, and leverage. With T formal out-of-sample months, annualised gross and net returns are defined as

$$R^{\text{gross,ann}} = 12 \left(\frac{1}{T} \sum_{t=1}^T R_t^{\text{gross}} \right),$$

and

$$R^{\text{net,ann}} = 12 \left(\frac{1}{T} \sum_{t=1}^T R_t^{\text{net}} \right).$$

Annualised trading cost is similarly

$$TC^{\text{ann}} = 12 \left(\frac{1}{T} \sum_{t=1}^T TC_t \right).$$

Let $s(R)$ denote the sample standard deviation of a monthly return series. Annualised volatility is

$$\sigma^{\text{ann}} = \sqrt{12} s(R),$$

and the corresponding Sharpe ratio for the excess-return series is

$$SR = \frac{R^{\text{ann}}}{\sigma^{\text{ann}}}.$$

Gross and net Sharpe ratios are computed from their respective return series.

The principal welfare measure is the monthly utility flow introduced in Section 3.1. Its annualised value is

$$U^{\text{ann}} = 12 \left(\frac{1}{T} \sum_{t=1}^T U_t \right).$$

This is the utility measure used in the main performance comparison.

Monthly turnover includes both trades among continuing securities and the liquidation of names leaving the universe:

$$TO_t = \sum_{i \in I_t} |\pi_{i,t} - h_{i,t}| + \sum_{i \in D_t} |h_{i,t}|.$$

Gross leverage is

$$Lev_t = \sum_{i \in I_t} |\pi_{i,t}|.$$

Reported turnover and leverage are their monthly sample averages.

For risk-calibration analysis, predicted annual volatility is

$$\sigma^{\text{pred}} = \sqrt{12 \left(\frac{1}{T} \sum_{t=1}^T \pi_t' \Sigma_t \pi_t \right)},$$

while realised annual volatility is

$$\sigma^{\text{real}} = \sqrt{12} s(R^{\text{net}}).$$

Their ratio,

$$\frac{\sigma^{\text{real}}}{\sigma^{\text{pred}}},$$

is used to assess whether the covariance model and portfolio scaling produce risks of the intended magnitude out of sample.

3.9.1 Inference on Utility Differences

The main statistical comparisons are conducted on monthly differences in utility flow. For methods A and B , define

$$\Delta U_t^{A,B} = U_t^A - U_t^B.$$

The annualised average difference is

$$\Delta U_{\text{ann}}^{A,B} = 12 \left(\frac{1}{T} \sum_{t=1}^T \Delta U_t^{A,B} \right).$$

The formal pairwise comparisons are Portfolio-ML Linear against Static-ML and Static-ML*, Portfolio-ML Random Features against the same two Static benchmarks, and Portfolio-ML Random Features against Portfolio-ML Linear. These comparisons keep the economic criterion fixed while varying the portfolio-construction method.

Serial dependence in monthly utility differences is addressed using the heteroskedasticity and autocorrelation consistent estimator of [14], with six monthly lags. Let

$$x_t = \Delta U_t - \overline{\Delta U},$$

and define

$$\hat{\Gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T x_t x_{t-\ell}.$$

The estimated long-run variance is

$$\hat{\Omega} = \hat{\Gamma}_0 + 2 \sum_{\ell=1}^6 \left(1 - \frac{\ell}{7}\right) \hat{\Gamma}_\ell.$$

Hence

$$SE(\overline{\Delta U}) = \sqrt{\frac{\hat{\Omega}}{T}},$$

and the HAC test statistic is

$$t_{\text{HAC}} = \frac{\overline{\Delta U}}{SE(\overline{\Delta U})}.$$

A circular block bootstrap following [15] provides a complementary finite-sample procedure while preserving short-run dependence in the monthly series. Blocks have length 12 months, and 2,000 bootstrap samples are drawn. For each replication, block starting points are sampled with replacement, the resulting blocks are concatenated, and the first T observations are retained. The 2.5th and 97.5th percentiles of the annualised bootstrap mean differences form the reported 95% confidence interval.

The bootstrap distribution is also used to report

$$P(\Delta U > 0),$$

defined as the fraction of bootstrap replications for which the mean utility difference is positive. Finally, monthly utility differences are aggregated within each calendar year to record the number of years in which one method delivers higher utility than another. Together, the HAC statistic, block-bootstrap interval, bootstrap probability, and year-by-year comparison distinguish a persistent economic advantage from one driven by a small number of isolated months.

The empirical design separates three levels of evaluation. Return prediction is judged by genuinely pre-test prediction error, portfolio hyperparameters are selected from accumulated out-of-sample portfolio histories, and the final comparison uses 2005–2024 returns net of implementation costs and adjusted for the risk chosen by each method. The empirical results that follow apply these conventions throughout. Additional implementation details, hyperparameter grids, and reproducibility checks are provided in Appendix A.

Chapter 4

Empirical Results

This chapter reports the formal out-of-sample results for the locked Top500/Core10 specification. The evaluation period runs from January 2005 to December 2024 and contains 240 monthly observations. The baseline investor has risk-aversion coefficient $\gamma = 10$ and fixed assets under management of USD 1 billion.

The central comparison is between prediction-first methods and Portfolio-ML. Return-ML Rank uses predicted-return rankings directly, Static-ML and Static-ML* combine predicted returns with a separate portfolio-optimisation stage, and Portfolio-ML learns the mapping from characteristics to an aim portfolio, with actual positions obtained through dynamic adjustment from inherited holdings. All methods are evaluated under the common return, transaction-cost, and ex-ante risk accounting developed in the previous chapter.

4.1 Baseline Out-of-Sample Performance

Table 4.1 reports the full-sample out-of-sample performance of the five formal methods.

Table 4.1: Baseline out-of-sample performance, 2005–2024

Method	Gross	Cost	Net	Vol.	SR _{net}	Utility	Turnover	Lev.
Return-ML Rank	0.16%	1.28%	-1.12%	6.29%	-0.179	-3.55%	46.32%	1.000
Static-ML	0.87%	0.10%	0.77%	3.76%	0.205	0.41%	13.76%	0.939
Static-ML*	0.93%	0.06%	0.87%	3.93%	0.222	0.46%	11.23%	0.730
Portfolio-ML Linear	3.15%	0.01%	3.14%	4.23%	0.743	2.10%	4.74%	0.389
Portfolio-ML RF	3.90%	0.04%	3.86%	5.21%	0.742	2.36%	11.42%	0.709

The table shows a clear progression as more of the investor’s economic problem enters portfolio construction. Return-ML Rank earns a small positive gross return, but its 46.32% average monthly turnover produces annualised trading costs of 1.28%, leaving a net return of -1.12% . Introducing covariance risk, inherited holdings, and transaction costs at the portfolio stage changes the outcome substantially. Static-ML and Static-ML* both produce positive net returns and utility while reducing turnover to 13.76% and 11.23%, respectively.

The largest performance gap in the baseline comparison appears between the prediction-first benchmarks and Portfolio-ML. Linear Portfolio-ML earns an annualised net return of 3.14% and utility of 2.10%; Random Features earns 3.86% and 2.36%. Their net Sharpe ratios are both approximately 0.74, compared with 0.22 for Static-ML*. Random Features produces the higher return point estimate, but with greater volatility, turnover, leverage, and trading cost, leaving its net Sharpe ratio almost identical to that of Linear. The main

baseline result is the gap between prediction-first portfolio construction and the Portfolio-ML architecture. The additional nonlinear flexibility of Random Features is secondary to that comparison.

4.2 Prediction, Trading Costs, and Implementability

The Return-ML Rank benchmark illustrates why predictive information and an implementable portfolio are different economic objects. Its annualised gross return is 0.16%, but average monthly turnover of 46.32% generates an annualised cost of 1.28%. The full-sample trading cost is approximately 8.23 times the absolute gross return, so the ranking signal does not generate enough gross performance to support this high-turnover implementation.

Figure 4.1 shows the same pattern over time. The separation between cumulative gross and net performance is largest for Return-ML Rank and much smaller for the transaction-cost-aware methods, especially Portfolio-ML.

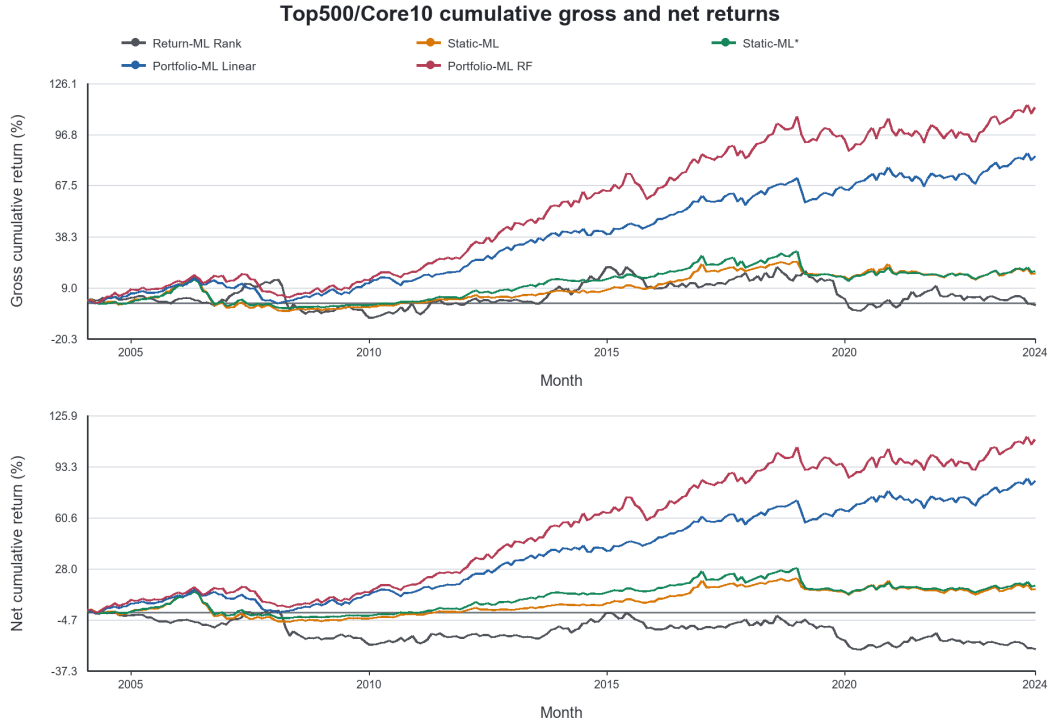


Figure 4.1: Cumulative gross and net portfolio performance, 2005–2024. The figure compares the five formal methods under the locked Top500/Core10 specification. Gross performance is shown before modelled transaction costs, while net performance deducts the common quadratic trading-cost measure.

Static portfolio optimisation recovers much of the implementation loss. Static-ML reduces turnover from 46.32% to 13.76%, lowers annualised cost from 1.28% to 0.10%, and raises net return from -1.12% to 0.77% . Static-ML* strengthens the same benchmark by allowing the portfolio stage to tune the scale of predicted returns, covariance risk, and perceived transaction costs. Turnover falls further to 11.23%, leverage to 0.730, and annualised trading cost to 0.06%, while utility rises from 0.41% to 0.46%.

This progression provides a strong prediction-first implementation benchmark before

Portfolio-ML is introduced. Static-ML* uses the same characteristics, covariance estimates, inherited portfolio state, and transaction-cost environment as Portfolio-ML, while allowing the portfolio stage to tune the scale of predicted returns, covariance risk, and perceived trading costs. Portfolio-ML additionally changes the architecture of the decision rule by estimating the characteristic-to-portfolio mapping directly and by embedding that mapping in the dynamic aim-and-adjustment structure. The comparison therefore measures the performance of these two portfolio architectures under a common implementation environment.

4.3 Portfolio-ML versus Prediction-First Portfolio Choice

Table 4.2 reports formal inference on monthly utility-flow differences.

Table 4.2: Out-of-sample utility differences

Comparison	ΔU	HAC t	Bootstrap 95% CI	$P(\Delta U > 0)$	Years won
Linear vs Static-ML	1.70%	1.928	[0.12%, 3.28%]	0.982	15/20
Linear vs Static-ML*	1.64%	2.086	[0.20%, 3.06%]	0.985	15/20
RF vs Static-ML	1.96%	1.911	[-0.06%, 3.87%]	0.968	15/20
RF vs Static-ML*	1.90%	2.015	[0.09%, 3.63%]	0.981	15/20
RF vs Linear	0.26%	0.417	[-1.01%, 1.37%]	0.670	13/20

Relative to Static-ML*, Linear Portfolio-ML increases annualised utility by 1.64 percentage points. The Newey–West statistic is 2.09, the 95% circular block-bootstrap interval is

$$[0.20\%, 3.06\%],$$

and the bootstrap probability of a positive mean difference is 98.5%. Random-Features Portfolio-ML increases annualised utility by 1.90 percentage points relative to Static-ML*, with a HAC statistic of 2.02, a bootstrap interval of

$$[0.09\%, 3.63\%],$$

and a 98.1% bootstrap probability of a positive mean difference. Both specifications also record higher annual utility than Static-ML* in 15 of the 20 formal test years.

The comparison with the untuned Static-ML benchmark is similar in magnitude. Linear improves annualised utility by 1.70 percentage points and Random Features by 1.96 points. The corresponding bootstrap interval for Linear lies above zero, while the Random-Features interval is $[-0.06\%, 3.87\%]$. The stronger and more informative comparison is with Static-ML*, because it gives the prediction-first approach an additional tuning layer while preserving the same information, risk, cost, and portfolio-accounting environment.

The economic result is direct. Improving return forecasts and then optimising a portfolio is not equivalent to estimating the portfolio mapping under the investor’s final objective. Under the common implementation assumptions used here, both Portfolio-ML specifications deliver materially higher out-of-sample utility than the strongest prediction-first benchmark.

The next chapter examines where this difference appears in gross returns, realised costs, trading intensity, inherited-position adjustment, and risk scaling. The remainder of this chapter asks how much additional value comes from nonlinear flexibility and whether the main ranking persists through time and across risk preferences.

4.4 Linear versus Random Features and Stability Through Time

Random Features produces the highest net-return point estimate in the baseline experiment, but Linear Portfolio-ML captures most of the economic gain over prediction-first portfolio choice. Relative to Linear, Random Features raises annualised net return by 0.73 percentage points and utility by 0.26 percentage points, while annualised volatility is almost one percentage point higher, average monthly turnover is 6.68 percentage points higher, and average gross leverage rises by approximately 0.32. The resulting net Sharpe ratio is marginally lower than that of Linear.

The utility difference between the two Portfolio-ML specifications is 0.26%, with

$$t_{\text{HAC}} = 0.42$$

and a bootstrap 95% interval of

$$[-1.01\%, 1.37\%].$$

The nonlinear specification therefore raises the return point estimate without a statistically distinguishable increase in utility. This makes the common Portfolio-ML architecture, rather than additional nonlinear model complexity, the more important empirical distinction in the baseline comparison.

The annual evidence supports the same ranking over much of the sample. Both Portfolio-ML specifications deliver higher annual utility than Static-ML* in 15 of the 20 formal test years. Figure 4.2 shows the annual distribution.

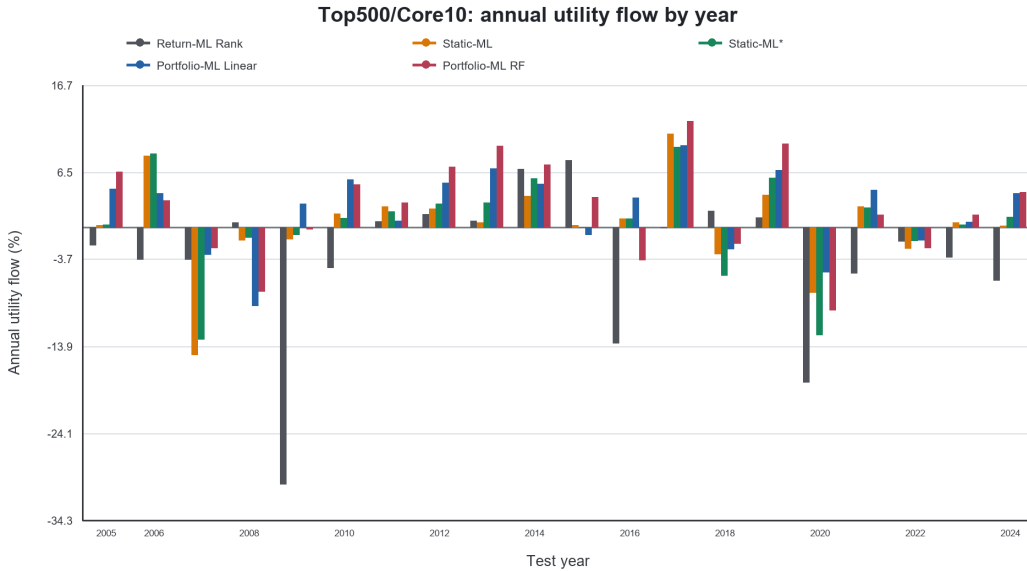


Figure 4.2: Annual utility flow by portfolio method, 2005–2024. Utility is measured using the common objective-consistent return, trading-cost, and ex-ante risk accounting.

The size of the relative contribution varies materially by year. The largest positive contribution occurs in 2007, which accounts for 20.6% of the sum of positive annual utility contributions for Linear and 19.0% for Random Features. The five largest positive years account for 62.6% and 56.5%, respectively. At the same time, 2008 is the largest negative relative contribution for both specifications: Linear trails Static-ML* by approximately 8.0 percentage points of annual utility and Random Features by 6.4 points.

The pattern is therefore not specific to crisis performance. A strong 2007 is followed by a weak 2008, while Portfolio-ML still wins in three quarters of the formal test years. The annual results point to a broad but uneven advantage, with a small number of years contributing disproportionately to its magnitude. Chapter 6 examines the contribution of 2007 more directly.

4.5 Risk-Aversion Sensitivity

The final sensitivity analysis varies the investor’s risk aversion. The full cumulative out-of-sample experiment is repeated over

$$\gamma \in \{2.5, 5, 7.5, 10, 15, 20, 30, 50\}.$$

The locked baseline remains $\gamma = 10$; the remaining values trace the performance of each method over a broader range of portfolio risk.

Across all eight values of risk aversion, both Portfolio-ML specifications deliver higher annualised utility than both Static benchmarks at the same γ .

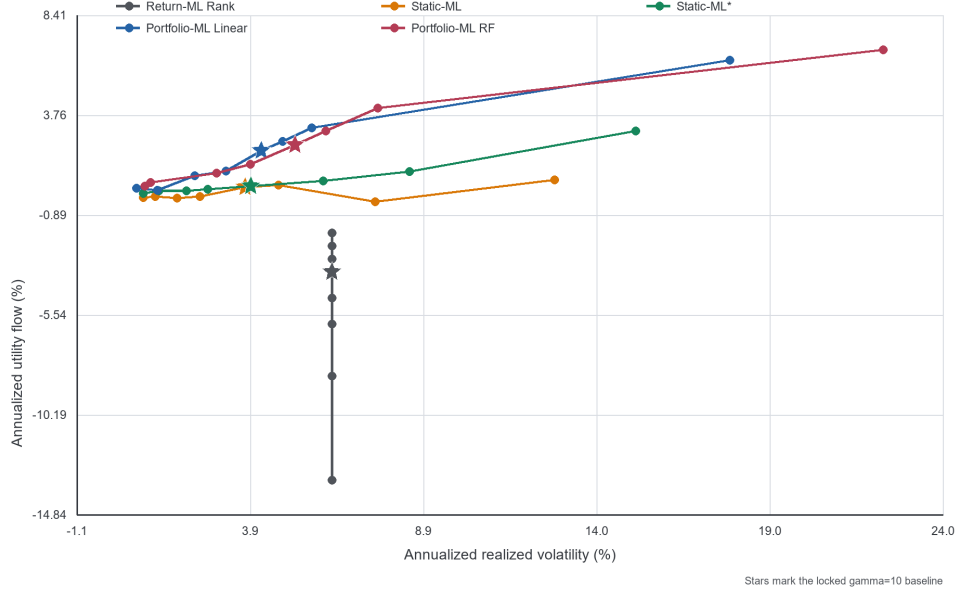


Figure 4.3: Utility and realised risk across risk-aversion parameters. Each point corresponds to a controlled cumulative out-of-sample specification at a fixed value of γ , with $\gamma = 10$ corresponding to the prespecified baseline.

At $\gamma = 2.5$, annualised utility is 6.32% for Linear and 6.80% for Random Features, compared with 0.75% for Static-ML and 3.02% for Static-ML*. At $\gamma = 20$, the corresponding utilities are 0.93%, 1.07%, -0.10% , and 0.24% . Realised volatility also declines monotonically as risk aversion increases for both Portfolio-ML specifications.

The risk-aversion exercise reinforces the baseline comparison. Portfolio-ML remains ahead across a broad range of investor risk preferences rather than only at $\gamma = 10$. The points are generated on a common discrete γ grid, so the exercise compares methods at the same risk aversion rather than interpolating them to exactly matched realised volatility. Within that design, the ordering is stable throughout the grid.

Overall, the Portfolio-ML architecture produces substantially higher implementable utility than prediction-first portfolio construction, with most of the gain already captured by the linear specification. This ranking persists across the tested risk preferences; the following chapters examine its implementation and robustness in greater detail.

Chapter 5

Economic Mechanisms and Implementation

The previous chapter established that Portfolio-ML delivers higher out-of-sample utility than the strongest prediction-first benchmark. This chapter examines how that advantage appears in the implemented portfolios. The main accounting difference is in gross return rather than realised transaction-cost savings, but the portfolio rules also differ materially in trading intensity, inherited-position adjustment, risk scaling, and their response to investor size. These dimensions provide a direct view of how portfolio-level learning changes the path from signals to implemented positions.

5.1 Gross Returns and Realised Trading Costs

Because Portfolio-ML internalises implementation costs in the learning problem, a natural explanation for its higher net performance would be lower realised trading costs. The accounting decomposition shows a different pattern: most of the net-return advantage is already present before realised costs are deducted.

Table 5.1 decomposes the annualised net-return difference between Portfolio-ML and the two Static benchmarks.

Table 5.1: Gross-return and trading-cost decomposition

Comparison	Gross-return diff.	Cost saving	Net-return diff.	Cost-saving share
Linear vs Static-ML	2.28%	0.09%	2.37%	3.77%
Linear vs Static-ML*	2.22%	0.05%	2.27%	2.00%
RF vs Static-ML	3.03%	0.06%	3.09%	2.04%
RF vs Static-ML*	2.97%	0.02%	2.99%	0.64%

Relative to Static-ML*, Linear Portfolio-ML earns 2.22 percentage points more annualised gross return, while the reduction in realised trading cost is 0.05 percentage points. Cost savings account for approximately 2% of the 2.27 percentage-point net-return difference. For Random Features, the gross-return difference is 2.97 percentage points and the cost saving is 0.02 percentage points, so less than 1% of the net-return difference comes from lower realised cost. The comparison with Static-ML shows the same pattern.

The accounting evidence shifts the economic question away from realised cost savings. Portfolio-ML pays somewhat less to trade, but the dominant difference is the gross return generated by the portfolio rules. The remaining sections examine the implementation choices that accompany that difference.

5.2 Trading Intensity, Portfolio Scale, and Liquidity

The two Portfolio-ML specifications reach their higher utility with different trading profiles. Table 5.2 compares Static-ML* with Linear and Random Features.

Table 5.2: Trading intensity and liquidity exposure

Method	Turnover	Leverage	Weighted ADV	Low-ADV	High-ADV
Static-ML*	11.23%	0.730	\$347.5m	4.21%	17.42%
Portfolio-ML Linear	4.74%	0.389	\$243.9m	10.90%	10.22%
Portfolio-ML RF	11.42%	0.709	\$303.4m	7.10%	14.03%

Linear Portfolio-ML has average monthly turnover of 4.74%, compared with 11.23% for Static-ML*, and its gross leverage is also materially lower. Random Features is different: turnover is 11.42%, slightly above Static-ML*, and leverage of 0.709 is close to the Static-ML* value of 0.730. Despite these differences, both Portfolio-ML specifications deliver substantially higher gross return and low realised trading cost. Low turnover is an important feature of the Linear strategy, but not a general explanation for the Portfolio-ML result.

The liquidity composition points in the same direction. The median absolute-weighted ADV of the Linear portfolio is approximately \$244 million, below the \$348 million of Static-ML*. Linear allocates 10.9% of absolute exposure to the lowest-ADV decile, compared with 4.2% for Static-ML*. Random Features also has lower weighted ADV and greater low-ADV exposure than Static-ML*, although the differences are smaller. Portfolio-ML is not obtaining its lower realised cost simply by concentrating in the most liquid securities. In the Linear case, much of the reduction in cost comes from trading less and carrying smaller positions even while holding more exposure to low-ADV names.

Current-universe costs are also less concentrated. The 10% of current names with the highest stock-level costs account for approximately 54.2% of current-universe cost for Static-ML*, compared with 48.7% for Linear and 45.3% for Random Features. This statistic excludes liquidation costs for securities leaving the universe, which are relatively important for Portfolio-ML, particularly Linear. It describes the distribution of costs among continuing holdings, not the concentration of total realised cost. Across the broader implementation profile, position size, inherited holdings, security selection, covariance risk, and the transaction-cost penalty all enter jointly.

5.3 Dynamic Adjustment towards the Aim Portfolio

Portfolio-ML separates the contemporaneous aim portfolio from the position that is actually implemented. With transaction costs, inherited holdings retain economic value whenever the benefit of immediate rebalancing is insufficient to justify the required trade. The size of this partial adjustment can be measured directly by comparing the implemented monthly trade with the trade required to move fully from grown inherited holdings to the contemporaneous aim.

Table 5.3 reports the corresponding distances.

Table 5.3: Adjustment from inherited holdings towards the aim

Method	Actual trade L1	Aim trade L1	Actual/Aim L1	Actual/Aim Λ distance
Portfolio-ML Linear	0.0438	0.1551	0.285	0.137
Portfolio-ML RF	0.1068	0.4141	0.264	0.169

The reported ratios are averages of the monthly ratios rather than ratios of full-sample average distances. For Linear Portfolio-ML, the implemented trade covers on average 28.5% of the immediate L1 distance from inherited holdings to the aim; the corresponding value for Random Features is 26.4%. The adjustment is even smaller in the transaction-cost-weighted distance

$$\sqrt{\Delta' \Lambda_t \Delta},$$

where the implemented trade covers 13.7% of the immediate-aim distance for Linear and 16.9% for Random Features.

Figure 5.1 illustrates the same result.

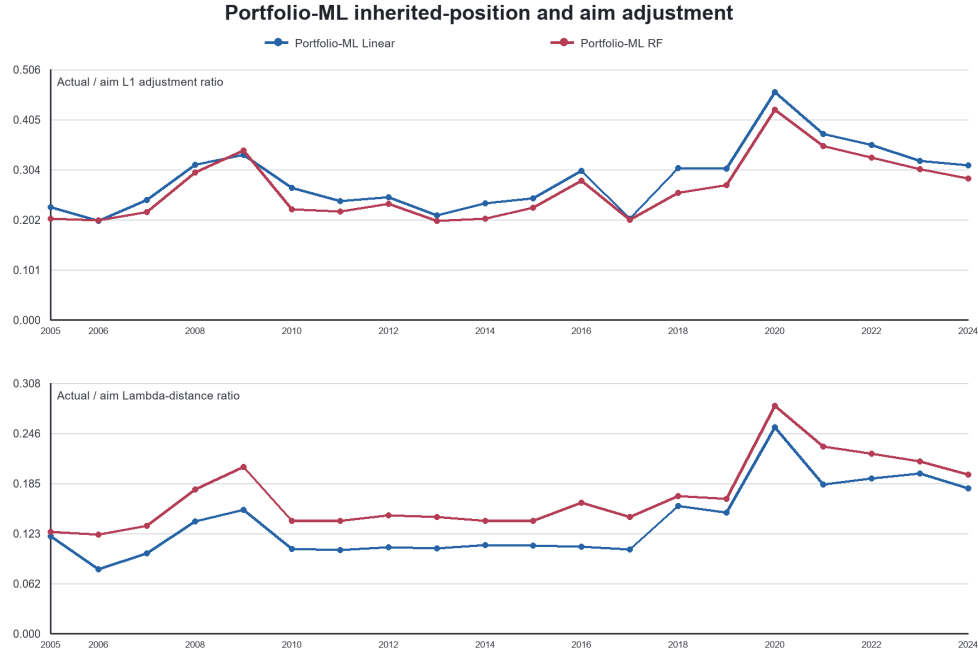


Figure 5.1: Adjustment from grown inherited holdings towards the Portfolio-ML aim portfolio. The implemented monthly trade is compared with the adjustment required to move immediately to the contemporaneous aim.

The contemporaneous aim acts as a direction of adjustment, not a position that is reached immediately. Most of the inherited portfolio remains in place each month, especially when distance is weighted by the same transaction-cost matrix used in the portfolio objective. This is the implementation pattern implied by gradual, cost-aware adjustment in [9] and [11]. The exercise describes the behaviour of the fitted Portfolio-ML recursion; it is not an independent structural estimation of the Gârleanu–Pedersen model.

5.4 Risk Scaling

A further difference appears in portfolio scale. When expected returns and covariance estimates are passed through a separate optimisation stage, estimation error can translate into large changes in position size. Direct portfolio-level learning may instead produce portfolios whose realised risk remains closer to the scale anticipated at the decision date.

Table 5.4 compares realised volatility with the volatility predicted by the common ex-ante covariance model.

Table 5.4: Realised-to-predicted volatility calibration

Method	Full sample	Median	P10	P90	Maximum
Static-ML	1.393	1.201	0.927	1.921	2.690
Static-ML*	1.369	1.108	0.921	1.674	2.378
Portfolio-ML Linear	0.928	0.852	0.647	1.253	1.317
Portfolio-ML RF	0.951	0.928	0.700	1.237	1.600

The Portfolio-ML ratios are close to one over the full sample and have substantially tighter upper tails. Linear has a full-sample realised-to-predicted ratio of 0.93 and Random Features 0.95, and neither specification records an annual ratio above two. The Static methods produce full-sample ratios of 1.39 for Static-ML and 1.37 for Static-ML*, with annual ratios above two in 2020 and 2021. The dedicated risk diagnostics attribute these episodes to underestimation of common-factor covariance during the pandemic and reopening periods rather than to covariance repair, fallback procedures, or exceptional leverage.

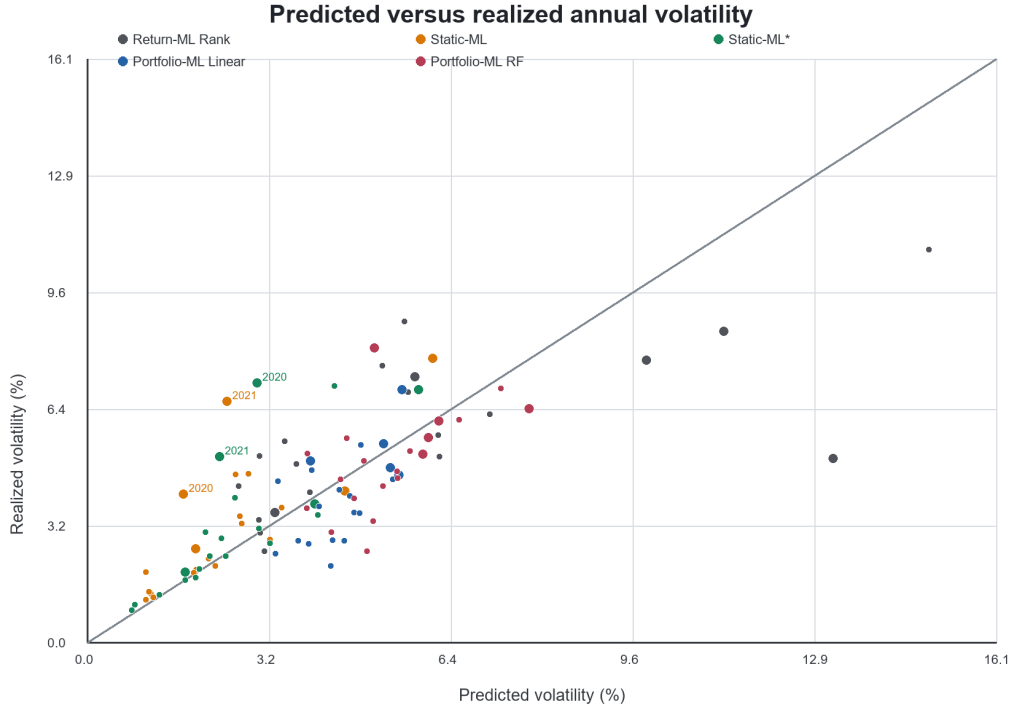


Figure 5.2: Predicted and realised annualised volatility. The diagonal represents exact ex-ante calibration.

Portfolio-ML keeps realised portfolio risk much closer to the common ex-ante scale in this sample. The result is strongest in the upper tail, where the Static methods experience large risk overshoots that are absent from the two Portfolio-ML specifications. This pattern is consistent with the economic argument in [11]: learning the portfolio rule directly changes not only which signals enter the portfolio, but also how their combined exposure is scaled.

5.5 AUM Sensitivity and Endogenous Adaptation

The baseline analysis fixes AUM at USD 1 billion. Because the quadratic transaction-cost term is proportional to investor size, varying AUM tests whether the portfolio rules adapt when implementation costs become more important.

The analysis considers fixed AUM levels of

$$0, \quad 0.25, \quad 0.5, \quad 1, \quad 2, \quad 5, \quad 10 \text{ billion USD.}$$

The zero-AUM point is a frictionless diagnostic. At every positive non-baseline AUM level, the portfolio experiment is recomputed under the same cumulative out-of-sample information protocol. Portfolio hyperparameters, Portfolio-ML coefficients, the dynamic adjustment matrix, portfolio state, and realised transaction costs are specific to the AUM scenario. The Return-ML prediction stream is shared because AUM does not enter the return-prediction model. Investor size changes the portfolio decision itself; the exercise is not a simple rescaling of realised costs on the baseline weights.

The clearest response is lower trading intensity. Linear Portfolio-ML turnover falls monotonically from 18.75% per month in the zero-cost diagnostic to 4.74% at USD 1 billion and 2.74% at USD 10 billion. Random-Features turnover falls from 69.28% to 11.42% and ultimately 5.36%. Gross leverage also declines with AUM for both specifications. At the same time, gross returns move down as increasingly aggressive trading opportunities become more expensive: Linear falls from 3.26% annually in the frictionless diagnostic to 3.14% at USD 1 billion and 2.70% at USD 10 billion, while Random Features falls from 4.54% to 3.90% and then 2.71%.

Both Portfolio-ML specifications retain positive net return and positive utility throughout the AUM grid. At USD 10 billion, Linear earns a net return of 2.66% and annualised utility of 1.82%, while Random Features earns 2.62% and 1.69%, respectively. These are model-implied implementability results under the adopted quadratic ADV specification, rather than external estimates of market capacity.

Figure 5.3 summarises the utility response.

Linear Portfolio-ML exceeds Static-ML* in utility at every AUM level tested. The utility gap is 3.23 percentage points in the zero-cost diagnostic, 1.64 points at USD 1 billion, and 1.36 points at USD 10 billion. The Random-Features gap is non-monotonic and also ends below its frictionless value. The main scalability result is the ability of Portfolio-ML to remain ahead while substantially changing its trading intensity and portfolio scale as implementation costs rise, even though the relative utility gap narrows.

The comparison between the two Portfolio-ML specifications also shifts at the upper end of the grid. Random Features has higher utility from zero through USD 5 billion, while Linear is marginally ahead at USD 10 billion by approximately 0.13 percentage points of annual utility. Linear also has lower turnover, leverage, and realised cost at every positive AUM level. The high-AUM comparison thus favours the simpler linear specification on implementation efficiency, while Random Features retains the higher utility point estimate over most of the tested range.

Selected hyperparameters show much less systematic movement than trading intensity. Linear Portfolio-ML selects the same ridge coefficient across the grid, while Random Features tends to move from a larger feature map in the frictionless case toward the smaller $p = 32$ specification as AUM increases. Most of the adaptation to investor size appears directly in the portfolio rule and dynamic adjustment; model complexity changes much less systematically.

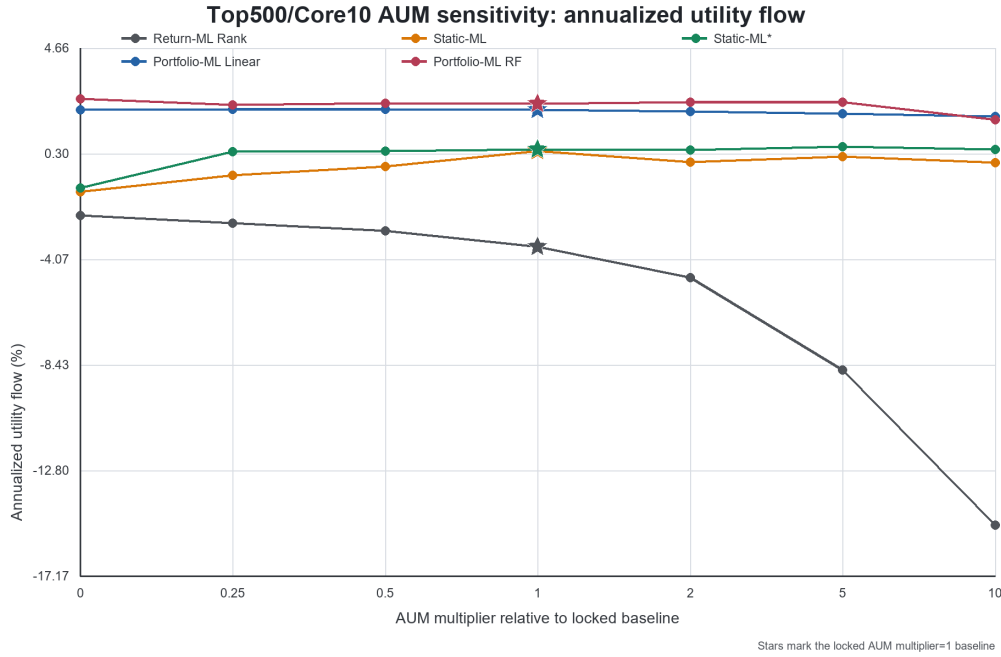


Figure 5.3: Annualised utility across fixed AUM levels. Each transaction-cost-aware point reflects an AUM-specific cumulative out-of-sample portfolio experiment. The USD 1 billion point is the locked baseline specification.

5.6 Economic Interpretation

The implementation evidence gives a coherent picture of where the Portfolio-ML advantage appears. Realised transaction-cost savings account for only a small share of the net-return difference, so the main accounting gain is already present in gross returns. The portfolio path, however, is materially different from the prediction-first benchmark. Linear trades and scales positions much less aggressively; Random Features shows that low turnover alone is not the explanation. Neither specification relies on a simple shift toward the most liquid securities. Both retain substantial inherited positions, move only part of the way toward the contemporaneous aim, and keep realised risk closer to the common ex-ante scale.

The AUM experiment strengthens this implementation-based interpretation. As market impact becomes more important, Portfolio-ML reduces turnover and leverage while remaining ahead of Static-ML* across the tested range. The common feature of these results is not any single implementation statistic, but the joint treatment of expected return, risk, trading costs, and inherited positions inside the portfolio rule. The evidence locates the Portfolio-ML advantage in this portfolio-level mapping, not in a single cost-saving or liquidity channel. Complete annual performance and sensitivity results are provided in Appendix B.

Chapter 6

Robustness and Signal Interpretation

The baseline results establish a clear comparison between learning approaches: Portfolio-ML delivers higher out-of-sample utility than the strongest prediction-first benchmark under Core10. This chapter asks how broadly that ranking survives and what the fitted portfolio policies reveal at the feature level. The first two exercises vary the information set and examine the contribution of 2007. The remaining sections measure economic feature importance and compare it with raw characteristic persistence and multi-horizon predictive persistence. Across these extensions, the portfolio-level ranking remains the clearest result, while the feature-level evidence provides a more nuanced account of which signals matter economically.

6.1 Robustness to the Information Set

Core10 uses a compact set of ten characteristics. Two richer specifications test whether the Portfolio-ML ranking survives a broader information set. Final15 adds five characteristics, while Final20 Conservative adds ten characteristics spanning additional momentum, profitability, value, safety, accrual, beta, and trading-activity information.

Final20 Conservative is the cleanest feature-set robustness exercise because its membership was fixed using information from 1995–2004. Selection used data quality, missingness, pre-OOS predictive content, persistence, redundancy, and theme coverage, without using formal 2005–2024 portfolio returns, realised utility, predictions, weights, selected hyperparameters, or final experiment outputs. Final15 is reported as a complementary richer-set comparison.

Table 6.1 compares the annualised utility of Static-ML* with the two Portfolio-ML specifications across the three information sets.

Table 6.1: Robustness across feature sets

Feature set	Static-ML*	Linear	RF	Linear-Static*	RF-Static*
Core10	0.46%	2.10%	2.36%	1.64%	1.90%
Final15	0.65%	2.18%	2.53%	1.53%	1.89%
Final20 Conservative	1.22%	2.16%	2.46%	0.95%	1.25%

The ranking is unchanged across all three information sets. Linear and Random-Features Portfolio-ML both deliver higher annualised utility than Static-ML*, with Linear-minus-Static* differences of 1.64, 1.53, and 0.95 percentage points for Core10, Final15, and Final20 Conservative. The corresponding Random-Features differences are 1.90, 1.89, and

1.25 percentage points. Random Features also retains the higher utility point estimate in every feature set.

Sampling uncertainty widens in the richer specifications. Under Core10, the block-bootstrap confidence intervals for both Portfolio-ML specifications relative to Static-ML* lie entirely above zero. For Final15, the Linear-minus-Static* utility difference has a HAC statistic of 0.89 and a bootstrap interval of approximately

$$[-1.06\%, 4.80\%],$$

while Random Features has a HAC statistic of 1.14 and an interval of approximately

$$[-0.70\%, 4.90\%].$$

For Final20 Conservative, the HAC statistics are 1.11 and 1.45 and both bootstrap intervals include zero. The feature-set result is strongest as a ranking result: Portfolio-ML remains ahead under both expansions, while the estimates are less precise than under Core10.

The richer sets also change the source of the relative advantage. Static-ML* utility rises from 0.46% under Core10 to 0.65% under Final15 and 1.22% under Final20 Conservative, whereas Portfolio-ML utility remains broadly stable around 2–2.5%. The richer information sets narrow the gap mainly by improving the prediction-first benchmark, while the Portfolio-ML ranking is unchanged. They also show that adding characteristics need not raise Portfolio-ML utility monotonically: the economic value of a broader information set depends on how the additional signals interact with covariance exposure, turnover, portfolio scale, and the existing characteristics.

6.2 Sensitivity to the 2007 Episode

The annual evidence shows that the Portfolio-ML advantage is distributed across the sample but uneven in magnitude. Both specifications outperform Static-ML* in annual utility in 15 of the 20 formal test years, while 2007 is the largest positive contributor. Under Core10, 2007 accounts for approximately 20.6% of total positive utility contributions for Linear and 19.0% for Random Features. An exclusion exercise provides a direct measure of how much this episode contributes to the full-sample evaluation.

The available analysis uses the completed Final15 experiment. The twelve realised observations from 2007 are removed and performance statistics, Newey–West inference, and the block bootstrap are recomputed over the remaining 228 months. The fitted models, selected hyperparameters, dynamic adjustment matrix, and post-2007 portfolio states are left unchanged. This is an evaluation-sensitivity exercise: it measures the contribution of the realised 2007 observations without reconstructing a new investment history in which 2007 is removed from estimation and portfolio-state evolution.

Table 6.2 reports the utility comparisons.

Table 6.2: Sensitivity to excluding 2007 from final evaluation

Comparison	Full sample	Excluding 2007	HAC t	Bootstrap 95% CI
Linear – Static-ML*	1.53%	0.00%	0.002	[-2.16%, 1.76%]
RF – Static-ML*	1.89%	0.51%	0.528	[-1.58%, 2.28%]

Removing the realised 2007 observations materially reduces the size of the estimated advantage. Linear falls from 1.53 percentage points to approximately zero, while Random Features falls from 1.89 to 0.51 percentage points. The ordering remains

$$\text{RF} > \text{Linear} > \text{Static-ML}^*,$$

but the remaining differences are estimated imprecisely, with HAC statistics below one and bootstrap intervals that include zero.

The result identifies 2007 as an important contributor to the magnitude of the full-sample utility gap. It is not, however, the sole source of the broader ranking: Portfolio-ML wins in 15 of 20 formal test years and remains ahead across all three feature sets. The combined evidence is best described as a broad ranking with concentrated magnitude. Because the exclusion leaves fitted models and subsequent portfolio states unchanged, it should be read as sensitivity of the realised evaluation rather than as a counterfactual no-2007 strategy.

6.3 Economic Feature Importance

The feature-set exercises show that the Portfolio-ML ranking extends beyond Core10. The next question is which characteristics matter most inside the fitted portfolio policies. Economic feature importance is measured by neutralising one characteristic while leaving the fitted policy and the rest of the economic environment unchanged.

For a fitted Portfolio-ML policy, feature j is neutralised by replacing its stored rank with the neutral value 0.5. The fitted annual coefficient vector, selected hyperparameters, Random-Features mapping where applicable, covariance inputs, transaction-cost inputs, risk aversion, AUM, and dynamic adjustment rule remain fixed. The portfolio is then re-evaluated recursively through the full formal sample, so neutralising a characteristic changes current positions, future inherited holdings, subsequent trades, exit liquidations, and realised utility. The resulting measure is

$$EFI_j = U^{\text{full}} - U^{\text{neutralised } j},$$

where both utilities are annualised under the common return-risk-cost accounting. Positive EFI means that neutralising the feature reduces realised utility for the fitted policy; negative EFI means that the neutralised path produces higher realised utility.

This is a fixed-model measure of economic dependence. Holding the fitted policy fixed isolates how much the realised portfolio path uses a characteristic under the estimated model. A remove-and-retrain exercise would answer a different question because coefficients, hyperparameters, feature substitution, and nonlinear interactions could all change.

Figure 6.1 reports the feature-level results.

Table 6.3 provides the corresponding values.

Table 6.3: Economic feature importance under the Core10 specification

Feature	Linear EFI	Rank	RF EFI	Rank
Momentum	0.216%	1	0.496%	1
One-month reversal	-0.004%	8	-0.016%	6
Volatility	0.071%	2	0.133%	2
Market beta	0.003%	5	0.016%	4
Idiosyncratic volatility	0.070%	3	-0.031%	8
Market capitalisation	-0.003%	7	-0.016%	5
Average dollar volume	0.002%	6	-0.024%	7
Book-to-market	-0.021%	10	-0.190%	10
Gross profitability	0.026%	4	0.088%	3
Asset growth	-0.021%	9	-0.036%	9

The strongest feature-level result is the agreement at the top of the ranking. Momentum has the largest positive EFI in both specifications, at approximately 0.22 percentage points for Linear and 0.50 percentage points for Random Features. Historical volatility

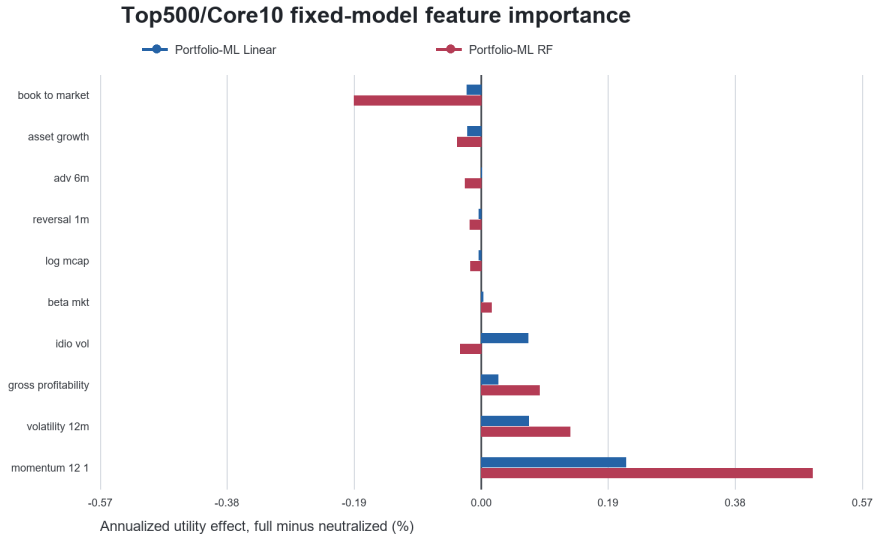


Figure 6.1: Fixed-model economic feature importance for Portfolio-ML Linear and Random Features. Importance is measured as the change in annualised realised utility when one rank characteristic is neutralised at 0.5 while fitted model parameters and economic inputs remain fixed.

ranks second in both, and gross profitability is positive and economically relevant under both specifications. Linear and Random Features agree on the sign of eight of the ten characteristics, and the rank correlation between their EFI rankings is approximately 0.78.

Book-to-market has the most negative EFI in both models and one-month reversal is mildly negative. These values mean that, conditional on the fitted multivariate policy, replacing the characteristic with a neutral input raises realised utility. In a joint model this can reflect redundancy, substitution, interaction with other signals, covariance exposures, or dynamic trading costs. The EFI ranking describes how the fitted policy uses the information set; it is distinct from a standalone ranking of univariate predictive quality.

The cross-model agreement on momentum and volatility is especially informative. Their economic importance emerges after predictive content is combined with portfolio scale, covariance, trading costs, inherited holdings, and the remaining characteristics. This motivates the next question: whether a simpler property of the signals can explain the EFI ranking.

6.4 Raw Characteristic Persistence and Economic Importance

Dynamic trading models give greater economic weight to return opportunities that persist long enough to justify gradual adjustment. The first persistence exercise asks whether the EFI ranking can be organised by a simple observable analogue: how slowly each Core10 characteristic rank changes over time.

For each adjacent pair of months, securities that remain in the dynamic Top500 universe are aligned and the cross-sectional Spearman correlation between the characteristic rank at t and the rank at $t + 1$ is calculated. The primary persistence statistic is the mean of these 239 monthly correlations. It measures cross-sectional state persistence, rather than stock-by-stock time-series autocorrelation.

The characteristics differ sharply on this measure. Momentum has mean monthly

persistence of approximately 0.89. Volatility, beta, idiosyncratic volatility, market capitalisation, ADV, book-to-market, gross profitability, and asset growth are all more persistent, with most values above 0.98. One-month reversal is almost non-persistent, with an average adjacent-month Spearman correlation close to zero.

Raw persistence provides little organisation of the EFI ranking. Across the ten Core10 characteristics, the Spearman correlation between persistence and Linear EFI is

$$0.212,$$

and the corresponding correlation for Random Features is

$$-0.091.$$

Leave-one-feature-out estimates range from zero to 0.55 for Linear and from approximately -0.27 to 0.17 for Random Features. A regression controlling for absolute Rank IC gives a standardised persistence coefficient close to zero in both specifications. With ten feature-level observations, the regression is best treated as a descriptive summary; the ranking itself carries the main economic message.

Momentum is the clearest counterexample to a simple slow-state story. It ranks only ninth in raw persistence but first in EFI under both models. Average dollar volume, market capitalisation, and book-to-market are among the most persistent characteristics, yet their fixed-model EFI is close to zero or negative. Book-to-market, for example, has monthly persistence of approximately 0.99 and negative EFI under both Linear and Random Features. Raw state persistence alone does not explain which characteristics matter most to the fitted portfolio policies.

This is economically plausible because state persistence and predictive persistence are different objects. A slowly moving characteristic need not retain the same relation with future returns, while a faster-moving state may encode a return opportunity that remains informative over several horizons. The next exercise measures persistence in the predictive relation itself.

6.5 Predictive Alpha Decay

Predictive persistence is measured directly from the multi-horizon relation between each Core10 characteristic and future returns. For each feature and each horizon

$$h = 1, \dots, 12,$$

the characteristic rank observed at month t is related to the single-month excess return realised at horizon $t + h$. The dependent variable is the incremental one-month return at each future horizon, not the cumulative return from t to $t + h$. Cross-sectional univariate regressions are estimated month by month and averaged using the Fama–MacBeth procedure [6], with Rank ICs calculated in parallel.

The sign of the one-month relationship defines the reference direction, and later-horizon coefficients are sign-adjusted relative to this initial direction. Signed future mass is the sum of the adjusted Fama–MacBeth coefficients from horizons two through twelve. The first decay horizon is the first point at which the sign-adjusted coefficient falls below one half of its initial magnitude, while the reversal horizon is the first horizon at which the adjusted coefficient changes sign. These are horizon descriptors rather than assumptions of monotonic decay, so a series may fall below the threshold and later recover.

Predictive persistence has more explanatory power for Random Features than raw state persistence, but the relation is not common across both Portfolio-ML specifications. The Spearman correlation between Fama–MacBeth signed future mass and Linear EFI is

$$0.115,$$

with a leave-one-feature-out range from approximately -0.10 to 0.38 . For Random Features, the corresponding correlation is

$$0.430,$$

and remains positive under leave-one-out perturbations. The RF result suggests that longer-lived univariate predictive content is associated with greater economic dependence in the nonlinear policy. Linear does not show the same pattern, and the alternative alpha-decay measures do not produce a stable cross-model ordering.

Figure 6.2 illustrates the Linear relation.

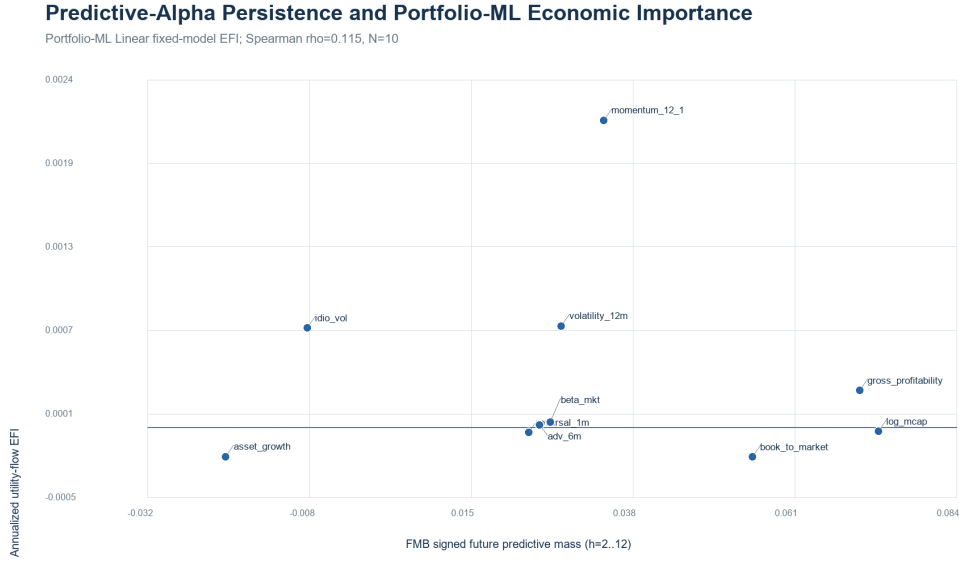


Figure 6.2: Predictive-alpha persistence and Linear Portfolio-ML EFI across the ten Core10 characteristics. Predictive persistence is measured using prespecified multi-horizon univariate return relationships. The cross-feature sample contains only ten observations and is interpreted descriptively.

Momentum shows why predictive persistence is more informative than raw state persistence. Its raw persistence is approximately 0.89, ranking ninth among the Core10 characteristics, but its one-month Fama–MacBeth slope is strongly positive at approximately 0.00696. Its signed future mass ranks fourth, the first half-decay occurs only at horizon six, and no opposite-direction Fama–MacBeth coefficient appears within the twelve-month window. Momentum combines a relatively fast-moving observed state with a more persistent predictive relation and the highest EFI in both Portfolio-ML specifications.

One-month reversal follows a different path. Its raw rank is essentially non-persistent; the one-month Fama–MacBeth slope is mildly positive while the one-month Rank IC is negative. The Fama–MacBeth path falls below half of its initial magnitude at horizon two, later recovers, and first crosses into the opposite direction at horizon seven. Its EFI is slightly negative in both Portfolio-ML specifications. The predictive path is non-monotonic: it weakens quickly, recovers, and later reverses.

Figure 6.3 compares the two horizon profiles.

The contrast extends beyond these examples. Book-to-market has raw persistence near 0.99 and does not reach the half-decay threshold within twelve months, yet its EFI is negative in both Portfolio-ML specifications. Average dollar volume and market capitalisation are also persistent states with near-zero or negative EFI. The evidence separates

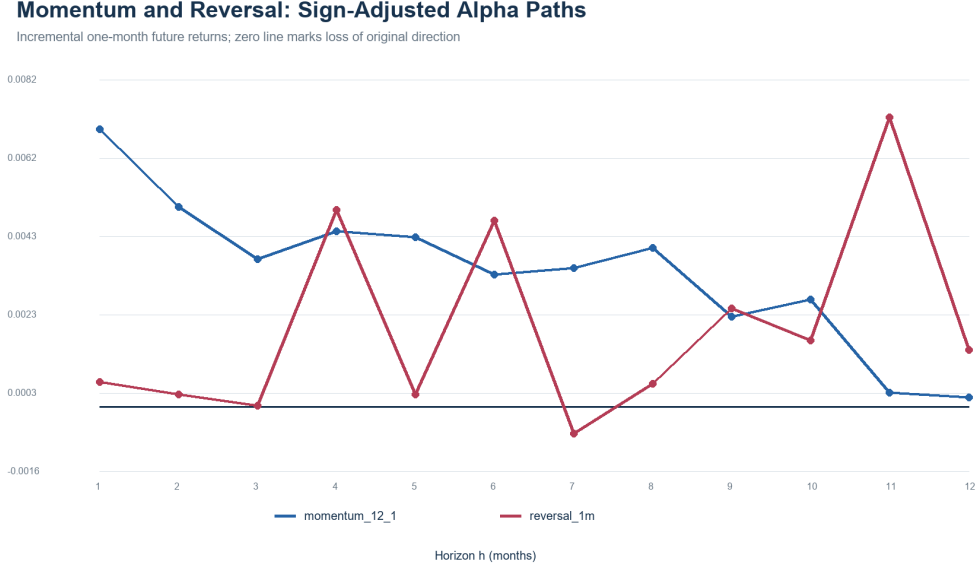


Figure 6.3: Multi-horizon predictive profiles for momentum and one-month reversal. The figure shows sign-adjusted Fama–MacBeth paths across horizons one to twelve.

three distinct objects:

state persistence, predictive persistence, economic portfolio value.

The first describes the evolution of the observed characteristic, the second its univariate relation with future returns, and the third the dependence of the fitted dynamic portfolio policy on that characteristic after interactions with the rest of the investment problem.

6.6 Relation to Dynamic Trading with Persistent Signals

The persistence exercises are motivated by [9]. In that framework, a return-predicting state is more valuable when the associated opportunity persists while the investor trades gradually toward the desired position. The same intuition helps motivate the dynamic structure of Portfolio-ML in [11].

The empirical objects here are observed characteristics and univariate future-return relations rather than estimated structural alpha states. The relevant question is whether the Gârleanu–Pedersen persistence intuition appears as a feature-level pattern inside the Core10 Portfolio-ML implementation. Four prespecified checks compare predictive persistence with raw persistence, cross-model consistency, leave-one-feature-out stability, and the momentum/reversal horizon profiles.

These checks do not produce a common monotone persistence–EFI relation. Random Features shows a positive association between signed future mass and EFI, while the Linear relation is weaker; the cross-metric signs are not uniform, and reversal has a non-monotonic horizon profile. The useful implication is that the persistence intuition does not map one-for-one from a structural return-predicting state to raw characteristic persistence or a single univariate alpha-decay statistic. Portfolio-ML uses the characteristics jointly, so predictive persistence remains one economic dimension among covariance exposure, interactions, trading costs, inherited holdings, and endogenous portfolio scale.

6.7 Interpretation

The robustness results preserve the main Portfolio-ML ranking across three information sets. Core10 provides the most precise inference, while Final15 and Final20 Conservative retain the same ordering with wider confidence intervals. The 2007 exercise shows that the magnitude of the estimated advantage is concentrated in part of the sample, but the broader annual evidence remains positive in 15 of 20 formal test years. Across both exercises, the central empirical comparison survives: Portfolio-ML remains ahead of the strongest prediction-first benchmark under the alternative evaluations considered here.

The feature-level analysis adds a second, more specific result. Momentum is the largest positive EFI feature in both Linear and Random Features, historical volatility ranks second, and the two EFI rankings have a cross-model correlation of approximately 0.78. Raw characteristic persistence explains little of this ordering. Multi-horizon predictive persistence is more informative for Random Features, but it does not produce the same relationship in Linear. Momentum captures the key distinction: it is not among the slowest-moving characteristics, yet its predictive content extends across several horizons and it is the most economically important feature in both fitted policies.

The comparison between learning approaches is clearer than any single feature-level explanation. Portfolio-ML continues to outperform the prediction-first benchmark across the richer information sets, while the EFI exercises show that economic importance is shaped by more than raw persistence or univariate alpha decay. The main result is a portfolio-level one: the Portfolio-ML architecture changes how predictive information is combined with covariance risk, trading costs, inherited positions, and portfolio scale. Additional robustness results and feature-level diagnostics are provided in Appendix C.

Conclusion

This dissertation studies whether aligning the machine-learning objective with the investor’s final portfolio objective improves implementable out-of-sample performance. Using a dynamic Top500 U.S. equity universe, common risk and transaction-cost inputs, and a cumulative out-of-sample design from 2005 to 2024, the analysis compares three prediction-first methods with Linear and Random-Features Portfolio-ML. The strongest prediction-first benchmark, Static-ML*, delivers an annualised net return of 0.87% and annualised utility of 0.46%, while Linear Portfolio-ML achieves a net return of 3.14% and utility of 2.10%, and Random-Features Portfolio-ML achieves 3.86% and 2.36%, respectively. The corresponding annualised utility improvements over Static-ML* are 1.64 percentage points for Linear and 1.90 percentage points for Random Features. The Newey–West statistics are 2.09 and 2.02, with circular block-bootstrap 95% confidence intervals of [0.20%, 3.06%] and [0.09%, 3.63%]. The main empirical result is therefore that the combined Portfolio-ML architecture delivers higher implementable utility than the strongest prediction-first alternative under the common implementation assumptions used throughout the dissertation.

The comparison between Linear and Random Features qualifies this result in an important way. Random Features produces the highest point estimate of net return, but its incremental utility gain over Linear is only 0.26 percentage points, with a HAC statistic of 0.42 and a bootstrap confidence interval of $[-1.01\%, 1.37\%]$. The two Portfolio-ML specifications also have almost identical net Sharpe ratios. Most of the gain over the prediction-first benchmarks is already captured by the linear Portfolio-ML specification. The evidence therefore indicates that the Portfolio-ML design matters more than the additional nonlinear flexibility of Random Features. This architecture combines portfolio-level estimation with dynamic aim-based adjustment, so the present comparison does not separately identify the contribution of each component.

The implementation results show that this advantage is not mainly a consequence of paying less to trade. Relative to Static-ML*, Linear Portfolio-ML earns 2.22 percentage points more annualised gross return but saves only about 0.05 percentage points of annualised trading cost; for Random Features, the cost-saving share of the net-return difference is below 1%. The dynamic diagnostics also show substantial partial adjustment: the implemented monthly trade covers on average 28.5% of the immediate L1 distance from inherited holdings to the Linear aim portfolio and 26.4% for Random Features. Realised-to-predicted volatility ratios are 0.93 and 0.95, compared with 1.37 for Static-ML*. These patterns are consistent with a portfolio rule that uses inherited positions, trading costs, and ex-ante risk jointly rather than repeatedly moving to a frictionless target. They do not, however, explain the source of the higher gross returns, which remains an important open question.

The AUM analysis shows that the portfolio rules respond materially as implementation costs become more severe. From the zero-cost diagnostic to USD 10 billion, average monthly turnover falls from 18.75% to 2.74% for Linear and from 69.28% to 5.36% for Random Features. Both specifications retain positive net returns and positive utility throughout the tested AUM range. At the same time, the relative advantage over Static-

ML* narrows with scale: for Linear, the utility gap declines from 3.23 percentage points in the zero-cost diagnostic to 1.36 percentage points at USD 10 billion. Portfolio-ML remains ahead across the AUM grid, although the advantage narrows with investor scale.

The robustness exercises lead to a similarly qualified conclusion. Portfolio-ML remains ahead of Static-ML* under both Final15 and Final20 Conservative, including the latter feature set constructed using only pre-formal-OOS information. The bootstrap intervals for these richer specifications include zero, however, so the statistical evidence is clearly strongest under Core10. The result is also uneven over time. Both Portfolio-ML specifications outperform Static-ML* in annual utility in 15 of the 20 formal test years, but 2007 contributes materially to the full-sample difference. In the available Final15 evaluation-only exclusion, removing the realised 2007 observations reduces the Linear utility advantage to approximately zero and the Random-Features advantage to 0.51 percentage points. The ranking survives these checks, but its magnitude and precision vary materially across information sets and periods.

At the feature level, the evidence does not show a stable relationship between persistence and economic importance. Momentum has the largest positive fixed-model EFI in both Portfolio-ML specifications, followed by historical volatility, while several more persistent characteristics have weak or negative EFI. Raw characteristic persistence is only weakly related to EFI, with cross-feature Spearman correlations of 0.21 for Linear and -0.09 for Random Features. Predictive-alpha persistence gives a somewhat stronger relationship for Random Features, with a correlation of 0.43, but only 0.12 for Linear and no stable pattern across the full set of prespecified diagnostics. Momentum is again the clearest counterexample to a simple persistence story: it ranks only ninth in raw state persistence but is the most important feature economically in both models and retains predictive content over several future horizons. These results suggest that persistence is only one part of the explanation. A characteristic's economic contribution is realised inside a multivariate portfolio problem in which predictive content interacts with covariance risk, trading costs, inherited holdings, and the remaining signals.

Taken as a whole, the evidence is much clearer about the comparison between learning approaches than about any single feature-level mechanism. Portfolio-ML consistently improves on the prediction-first benchmarks under the baseline specification and retains that ranking across the additional feature sets and AUM experiments. The evidence for why individual characteristics matter is less uniform, and neither lower realised trading costs, additional nonlinear flexibility, nor a single measure of signal persistence is sufficient to account for the full result.

Several extensions would help establish how broadly these conclusions apply. The most immediate is to study the source of the gross-return difference more directly, since this component explains most of the performance gap but is not identified by the current accounting decomposition. A second direction is a fully counterfactual temporal robustness exercise in which influential periods such as 2007 are removed from model estimation, validation, and subsequent portfolio-state evolution, so that the entire cumulative out-of-sample path is reconstructed. External validation in other equity universes, markets, and asset classes would also test whether the ranking survives under different liquidity conditions, covariance structures, and signal dynamics. Finally, a more structural model of expected-return dynamics could provide a closer empirical link between estimated alpha persistence and the dynamic trading states studied by [9].

The broader implication is that machine learning for investment should be evaluated through the portfolio decisions it generates. In this setting, portfolio-level learning with dynamic implementation materially improves out-of-sample utility, and the linear specification captures most of the gain. The key economic margin is how predictive information is translated into implementable portfolio positions.

Appendix A

Empirical Implementation and Reproducibility

This appendix records the implementation choices that are most useful for reproducing the empirical comparison. The main text develops the economic motivation and the portfolio equations; the material below fixes the numerical grids, timing conventions, dynamic-universe accounting, Random-Features construction, and final integrity checks used in the locked experiments.

A.1 Hyperparameter Grids and Fixed Settings

Table A.1 reports the candidate grids used in the formal cumulative out-of-sample procedure. Return-ML candidates are selected by pre-test prediction mean squared error. Static-ML and Static-ML* use the cumulative validation score based on realised gross returns, trading costs, and the population variance of realised gross returns. Portfolio-ML candidates are selected by cumulative realised monthly utility flow. The selection-frequency column records the choices made across the twenty formal test years and provides a compact check on the stability of the annual selection procedure.

Table A.1: Hyperparameter grids and formal-year selection frequencies

Method	Parameter	Candidate values	Selection frequency, 2005–2024
Return-ML Linear	Ridge λ	$\{0.01, 0.1, 1, 10, 100, 1000\}$	0.01: 3; 0.1: 1; 1: 1; 10: 4
Return-ML RF	Ridge λ	$\{0.01, 0.1, 1, 10, 100, 1000\}$	0.01: 4; 0.1: 1; 1000: 6
Return-ML RF	p	$\{32, 64\}$	32: 9; 64: 2
Return-ML RF	η	$\{0.1, 0.5\}$	0.1: 8; 0.5: 3
Static-ML	s	$\{0.01, 0.05, 0.1, 0.25, 0.5, 1\}$	0.1: 9; 0.25: 11
Static-ML	ϕ	$\{1, 3, 6, 12\}$	1: 18; 3: 1; 6: 1
Static-ML*	u	$\{0.05, 0.1, 0.25, 0.5, 1\}$	0.1: 5; 0.25: 12; 0.5: 3
Static-ML*	v	$\{0, 0.25, 0.5, 1\}$	0: 2; 0.5: 1; 1: 17
Static-ML*	k	$\{0.5, 1, 2\}$	0.5: 1; 1: 1; 2: 18
Portfolio-ML Linear	Ridge λ	$\{0.1, 1, 10, 100, 1000, 10000\}$	1000: 20
Portfolio-ML RF	Ridge λ	$\{0.1, 1, 10, 100, 1000, 10000\}$	1000: 20
Portfolio-ML RF	p	$\{32, 64\}$	32: 18; 64: 2
Portfolio-ML RF	η	$\{0.1, 0.5\}$	0.1: 12; 0.5: 8

Return-ML Linear is selected in 9 of 20 test years and Return-ML Random Features in 11 of 20; the RF-specific frequencies therefore sum to 11. Portfolio-ML RF frequencies refer to all twenty formal years.

The fixed numerical settings are collected in Table A.2. Baseline AUM is a scalar USD 1 billion and remains fixed through time. The portfolio state evolves dynamically,

while the cost-scaling wealth level does not compound with realised returns.

Table A.2: Fixed settings in the locked baseline implementation

Setting	Value	Application
Baseline risk aversion γ	10	portfolio choice and utility
Baseline AUM W	USD 1 billion	optimisation and realised costs
Random seed	20260710	RF maps and bootstrap
Matrix- m updates	10	full Portfolio-ML adjustment matrix
Expected monthly market return	0.007	matrix m
Monthly risk-free rate	0	matrix m
Static covariance ridge	10^{-8}	Static-ML and Static-ML* solves
Portfolio-ML volatility floor	0.01	design-matrix volatility scaling
Portfolio-ML ridge floor	10^{-10}	closed-form coefficient solve
ADV cost coefficient	0.2	$\lambda_{i,t} = 0.2/ADV_{i,t}^{6m}$
Covariance eigenvalue floor	10^{-6}	monthly stock covariance
Risk-regression ridge	0.001	daily factor-return regression
Risk EWMA span	60 days	factor and idiosyncratic variance
Daily-to-monthly multiplier	21	covariance conversion

A.2 Cumulative Out-of-Sample Timeline

The formal design is cumulative and respects the information set available before each test year. Return history begins in 1990, portfolio-level estimation begins in 1995, candidate portfolio validation begins in 1996, and the formal test spans January 2005 to December 2024. For a formal year Y , all estimation and selection information ends in November of $Y - 1$. December $Y - 1$ is excluded from the refit because its one-month-ahead return belongs to the next calendar month. The selected model is then executed from January to December of year Y , and the realised December portfolio state is carried into January $Y + 1$.

Table A.3: Timing of the cumulative out-of-sample design

Stage	Start	End	Role
Master return history	Jan. 1990	Dec. 2024	prediction history and risk warm-up
Portfolio-analysis history	Jan. 1995	Dec. 2024	portfolio fitting and evaluation frame
Candidate portfolio OOS	1996	$Y - 1$	cumulative pre- Y validation history
Formal OOS	Jan. 2005	Dec. 2024	locked 240-month evaluation
Annual refit cutoff	available history	Nov. $Y - 1$	prediction and Portfolio-ML refits
State continuity	Dec. $Y - 1$	Jan. Y	inherited portfolio carried forward

Within each formal year, Return-ML model class and hyperparameters are chosen from pre- Y prediction validation and the selected predictor is refitted through the November cutoff. The resulting prediction stream is shared by Return-ML Rank, Static-ML, and Static-ML*. Portfolio candidates use only validation years preceding Y ; Static methods select their decision parameters from the cumulative Static score, while Portfolio-ML selects the portfolio specification from cumulative realised validation utility and refits β through the same cutoff. No test-year outcome enters model selection or refitting, and the portfolio state is not reset at calendar-year boundaries.

A.3 Dynamic-Universe Accounting

The dynamic Top500 universe requires explicit treatment of continuing securities, entrants, and exits. For a continuing security, the previous position is grown by the realised stock return and normalised by the market-growth factor,

$$h_{i,t} = \frac{\pi_{i,t-1}(1 + r_{i,t})}{G_t}, \quad G_t = \max \left(1 + \sum_i \omega_{i,t-1}^{mkt} r_{i,t}, 0.1 \right),$$

where $\omega_{i,t-1}^{mkt}$ denotes the market-equity weights used by the implementation. The local field used for these weights is stored as `lag_mktcap`; the audited field corresponds to the same-row JKP market-equity variable `me`, with no additional local one-month shift. The normalisation affects inherited weights and does not alter the fixed scalar AUM.

For current-universe securities, $\Delta_t = \pi_t - h_t$. Exiting securities are compulsorily liquidated, with exit trade $\ell_t = -h_t$ on the dropped names. The realised monthly trading cost and turnover are

$$TC_t = \frac{W}{2} (\Delta'_t \Lambda_t \Delta_t + \ell'_t \Lambda_{t-1} \ell_t), \quad TO_t = \sum_{i \in I_t} |\Delta_{i,t}| + \sum_{i \in E_t} |\ell_{i,t}|.$$

The first formal portfolio in January 2005 is initialised with market-equity weights; the implementation contains market-capitalisation and equal-weight fallbacks for that initialisation. Thereafter the state evolves continuously through the full formal sample.

Table A.4: Accounting treatment of dynamic-universe membership

Status	Inherited position	Trade	Cost treatment
Continuing	grown prior weight	$\pi_{i,t} - h_{i,t}$	current $\lambda_{i,t}$
Entrant	0	$\pi_{i,t}$	current $\lambda_{i,t}$
Exit	grown prior weight	$-h_{i,t}$	previous stored $\lambda_{i,t-1}$

A.4 Random-Features Implementation

The two Random-Features applications use the same cosine–sine construction but different economic preprocessing. Return-ML RF applies the map directly to the rank-normalised inputs after filling non-finite values with 0.5,

$$\Phi_p(X) = \frac{1}{\sqrt{p}} [\cos(XW), \sin(XW)], \quad w_j \sim N(0, \eta^2 I), \quad j = 1, \dots, p/2,$$

so p is the final number of sinusoidal columns. An intercept is added after the transformation and is left unpenalised in the prediction Ridge regression. Return-ML RF does not apply the training-sample standardisation used by Return-ML Linear.

Portfolio-ML RF begins from the same rank representation and Random-Features map, then de-means each mapped column within month, normalises each column to unit cross-sectional Euclidean norm, adds a $1/\sqrt{N_t}$ constant, and scales each stock row by the stock volatility from the repaired covariance matrix, subject to the 0.01 floor. The complete coefficient vector, including the constant coefficient, is Ridge-penalised. With fixed η and seed, the $p = 32$ map is a prefix of the $p = 64$ Portfolio-ML projection bank.

Table A.5: Preprocessing and estimation across the four characteristic mappings

Model	Input	Standardisation / scaling	Feature map	Constant treatment
Return-ML Linear	ranks, missing = 0.5	training-sample mean and s.d.	linear	intercept, unpenalised
Return-ML RF	ranks, missing = 0.5	none before RF	cosine-sine	intercept, unpenalised
Portfolio-ML Linear	ranks, missing = 0.5	monthly de-mean, L2, volatility	linear	$1/\sqrt{N_t}$, penalised
Portfolio-ML RF	ranks, missing = 0.5	RF then monthly de-mean, L2, volatility	cosine-sine	$1/\sqrt{N_t}$, penalised

A.5 Implementation Integrity Checks

The final preflight and reconciliation diagnostics are summarised in Table A.6. They confirm the intended sample length, complete availability of repaired risk inputs, absence of factor or matrix- m fallbacks, and the timing checks used to guard against future-information contamination. The numerical reconciliation errors are several orders of magnitude below the corresponding tolerances.

Table A.6: Final implementation-integrity diagnostics

Diagnostic	Final result
Formal OOS months	240
Usable risk-model months	360/360
Factor fallback count	0
Covariance repair count	0
Matrix- m fallback count	0
Future-information violations	0
Validation-cutoff violations	0
Baseline metric recomputation max. abs. error	9.89×10^{-17}
Baseline workbook reconciliation max. abs. error	0
Utility-accounting max. abs. error	6.94×10^{-18}
Final preflight gate	PASS
Checkpoint/resume verification	PASS

Appendix B

Supplementary Performance and Sensitivity Results

This appendix provides the complete annual and sensitivity results that support the summary evidence reported in the main text. The tables retain the same accounting convention throughout: annual utility is the annualised ex-ante utility flow, turnover is average monthly absolute trading including exits, and leverage is the average gross absolute portfolio weight.

B.1 Annual Utility, 2005–2024

Table B.1 reports annual utility for all five formal methods. Portfolio-ML Linear and Random Features each exceed Static-ML* in 15 of the 20 formal years, while Random Features exceeds Linear in 13 years. The largest positive relative contribution for both Portfolio-ML specifications occurs in 2007, when all three cost-aware methods record negative absolute utility but the Portfolio-ML losses are substantially smaller than the Static-ML* loss. The largest negative relative contribution occurs in 2008. The 2007 observation accounts for 20.6% of Linear’s and 19.0% of Random Features’ summed positive annual utility differences relative to Static-ML*.

B.2 Annual Hyperparameter Selection

The annual selections in Table B.2 show that the portfolio-level regularisation choices are considerably more stable than the prediction model class. Portfolio-ML Linear selects $\lambda = 1000$ in all twenty years, and Portfolio-ML RF also selects $\lambda = 1000$ in every formal year. The table therefore reports the changing RF width and scale while omitting these two constant Ridge columns. Static-ML moves from $s = 0.10$ to $s = 0.25$ in the later sample and selects $\phi = 1$ in eighteen years; Static-ML* selects $k = 2$ in eighteen years and $v = 1$ in seventeen years.

B.3 Risk-Aversion Sensitivity

Table B.3 reports the full locked γ grid for Static-ML* and the two Portfolio-ML specifications. Each non-baseline point is a complete cumulative out-of-sample run under the same Top500/Core10 universe, risk model, transaction-cost model, and timing protocol. The $\gamma = 10$ rows reconcile exactly to the baseline. Portfolio-ML utility exceeds Static-ML* at every common value in the tested grid, while realised volatility and trading intensity

Table B.1: Annualised utility flow by formal test year (percentage points)

Year	Rank	Static	Static*	PML Linear	PML RF
2005	-2.07	0.29	0.35	4.58	6.58
2006	-3.75	8.45	8.65	4.05	3.16
2007	-3.73	-14.89	-13.06	-3.15	-2.35
2008	0.63	-1.49	-1.13	-9.14	-7.48
2009	-30.07	-1.33	-0.79	2.80	-0.14
2010	-4.71	1.63	1.12	5.64	5.07
2011	0.73	2.51	1.90	0.81	2.92
2012	1.58	2.20	2.83	5.23	7.11
2013	0.79	0.62	2.93	6.96	9.59
2014	6.90	3.70	5.80	5.10	7.41
2015	7.91	0.27	-0.00	-0.82	3.57
2016	-13.54	1.09	1.03	3.53	-3.78
2017	-0.05	11.01	9.45	9.65	12.48
2018	1.97	-3.10	-5.61	-2.49	-1.87
2019	1.18	3.85	5.86	6.76	9.84
2020	-18.16	-7.58	-12.58	-5.19	-9.64
2021	-5.34	2.47	2.36	4.41	1.49
2022	-1.56	-2.43	-1.53	-1.45	-2.35
2023	-3.44	0.61	0.38	0.70	1.50
2024	-6.20	0.25	1.22	4.05	4.17

Table B.2: Selected annual hyperparameters

Year	Return model	Return λ	Return RF (p, η)	Static (s, ϕ)	Static* (u, v, k)	PML RF (p, η)
2005	Linear	0.1	—	(0.1,6)	(0.1,0,0.5)	(32,0.1)
2006	Linear	0.01	—	(0.1,3)	(0.25,1,2)	(32,0.1)
2007	Linear	0.01	—	(0.25,1)	(0.25,0,1)	(32,0.1)
2008	Linear	0.01	—	(0.1,1)	(0.1,0.5,2)	(32,0.1)
2009	RF	0.01	(64,0.5)	(0.1,1)	(0.1,1,2)	(32,0.1)
2010	RF	1000	(32,0.1)	(0.1,1)	(0.1,1,2)	(32,0.1)
2011	RF	0.1	(64,0.5)	(0.1,1)	(0.1,1,2)	(32,0.1)
2012	Linear	10	—	(0.1,1)	(0.25,1,2)	(32,0.1)
2013	RF	1000	(32,0.1)	(0.1,1)	(0.25,1,2)	(32,0.1)
2014	RF	1000	(32,0.1)	(0.1,1)	(0.25,1,2)	(32,0.1)
2015	RF	0.01	(32,0.1)	(0.25,1)	(0.25,1,2)	(32,0.1)
2016	RF	0.01	(32,0.1)	(0.25,1)	(0.25,1,2)	(64,0.5)
2017	RF	1000	(32,0.1)	(0.25,1)	(0.25,1,2)	(32,0.1)
2018	RF	1000	(32,0.1)	(0.25,1)	(0.5,1,2)	(32,0.5)
2019	RF	1000	(32,0.1)	(0.25,1)	(0.5,1,2)	(32,0.5)
2020	Linear	10	—	(0.25,1)	(0.5,1,2)	(64,0.5)
2021	RF	0.01	(32,0.5)	(0.25,1)	(0.25,1,2)	(32,0.5)
2022	Linear	10	—	(0.25,1)	(0.25,1,2)	(32,0.5)
2023	Linear	1	—	(0.25,1)	(0.25,1,2)	(32,0.5)
2024	Linear	10	—	(0.25,1)	(0.25,1,2)	(32,0.5)

generally contract as risk aversion increases. These are common- γ comparisons; the table does not interpolate methods to matched realised volatility.

Table B.3: Risk-aversion sensitivity for the principal methods

γ	Method	Net return (% p.a.)	Volatility (% p.a.)	Utility (% p.a.)	Turnover (% monthly)
2.5	Static-ML*	4.57	15.10	3.02	33.51
2.5	PML Linear	10.21	17.83	6.32	15.40
2.5	PML RF	12.48	22.28	6.80	30.95
5	Static-ML*	2.13	8.54	1.12	21.51
5	PML Linear	4.10	5.69	3.17	4.98
5	PML RF	5.61	7.62	4.08	15.00
7.5	Static-ML*	1.42	6.02	0.69	16.57
7.5	PML Linear	3.56	4.85	2.53	4.86
7.5	PML RF	4.54	6.09	3.02	12.81
10	Static-ML*	0.87	3.93	0.46	11.23
10	PML Linear	3.14	4.23	2.10	4.74
10	PML RF	3.86	5.21	2.36	11.42
15	Static-ML*	0.60	2.68	0.31	9.14
15	PML Linear	2.01	3.21	1.15	4.51
15	PML RF	2.65	3.92	1.47	9.02
20	Static-ML*	0.46	2.05	0.24	8.03
20	PML Linear	1.46	2.29	0.93	3.21
20	PML RF	1.88	2.94	1.07	6.64
30	Static-ML*	0.37	1.25	0.24	5.74
30	PML Linear	0.48	1.21	0.26	2.06
30	PML RF	0.79	1.02	0.62	3.36
50	Static-ML*	0.20	0.80	0.12	4.15
50	PML Linear	0.47	0.61	0.36	1.20
50	PML RF	0.65	0.85	0.46	3.15

B.4 AUM Sensitivity

The AUM exercise uses fixed scalar investor sizes of USD 0, 0.25, 0.5, 1, 2, 5, and 10 billion. Each non-baseline scenario reconstructs the portfolio-validation history, portfolio hyperparameter selection, Portfolio-ML coefficient fit, adjustment matrix, state path, and realised transaction costs for that AUM. The Return-ML prediction stream is shared because AUM is not an input to the prediction problem. The zero-AUM case is the frictionless diagnostic in which the transaction-cost term vanishes.

The relative utility gaps are shown separately in Table B.5. Linear maintains a positive advantage over Static-ML* throughout the grid, although the gap declines from 3.23 percentage points in the frictionless diagnostic to 1.36 percentage points at USD 10 billion. Random Features remains ahead of Linear through USD 5 billion; at USD 10 billion the Linear-minus-RF utility difference is approximately 0.13 percentage points.

B.5 Annual Risk Calibration

Table B.6 reports the yearly ratio of realised to predicted annualised volatility. The Static methods exhibit the largest upper-tail misses, with ratios above two in 2020 and 2021. Portfolio-ML has no year above two; its maximum yearly ratios are 1.32 for Linear and 1.60 for Random Features. The full-sample ratios are 1.39 for Static-ML, 1.37 for Static-ML*, 0.93 for Linear, and 0.95 for Random Features.

Table B.4: AUM sensitivity for Static-ML* and Portfolio-ML

AUM (USD bn)	Method	Cost (% p.a.)	Net return (% p.a.)	Utility (% p.a.)	Turnover (% monthly)	Leverage
0	Static-ML*	0.00	-0.98	-1.13	49.20	0.654
0	PML Linear	0.00	3.26	2.10	18.75	0.419
0	PML RF	0.00	4.54	2.55	69.28	0.979
0.25	Static-ML*	0.05	0.80	0.38	18.98	0.910
0.25	PML Linear	0.00	3.23	2.14	6.16	0.403
0.25	PML RF	0.02	3.96	2.31	15.77	0.769
0.5	Static-ML*	0.07	0.86	0.40	15.71	0.908
0.5	PML Linear	0.01	3.20	2.13	5.42	0.397
0.5	PML RF	0.03	3.94	2.37	13.14	0.732
1	Static-ML*	0.06	0.87	0.46	11.23	0.730
1	PML Linear	0.01	3.14	2.10	4.74	0.389
1	PML RF	0.04	3.86	2.36	11.42	0.709
2	Static-ML*	0.08	0.95	0.45	9.74	0.695
2	PML Linear	0.02	3.04	2.05	4.10	0.378
2	PML RF	0.05	3.78	2.41	9.33	0.658
5	Static-ML*	0.06	0.96	0.58	5.95	0.519
5	PML Linear	0.03	2.84	1.94	3.30	0.357
5	PML RF	0.07	3.60	2.43	6.93	0.588
10	Static-ML*	0.07	0.78	0.47	5.05	0.442
10	PML Linear	0.04	2.66	1.82	2.74	0.338
10	PML RF	0.08	2.62	1.69	5.36	0.491

Table B.5: AUM sensitivity of annualised utility differences (percentage points)

AUM (USD bn)	Linear – Static*	RF – Static*	RF – Linear
0		3.68	0.45
0.25	1.75	1.92	0.17
0.5	1.73	1.97	0.25
1	1.64	1.90	0.26
2	1.60	1.97	0.37
5	1.36	1.85	0.49
10	1.36	1.23	-0.13

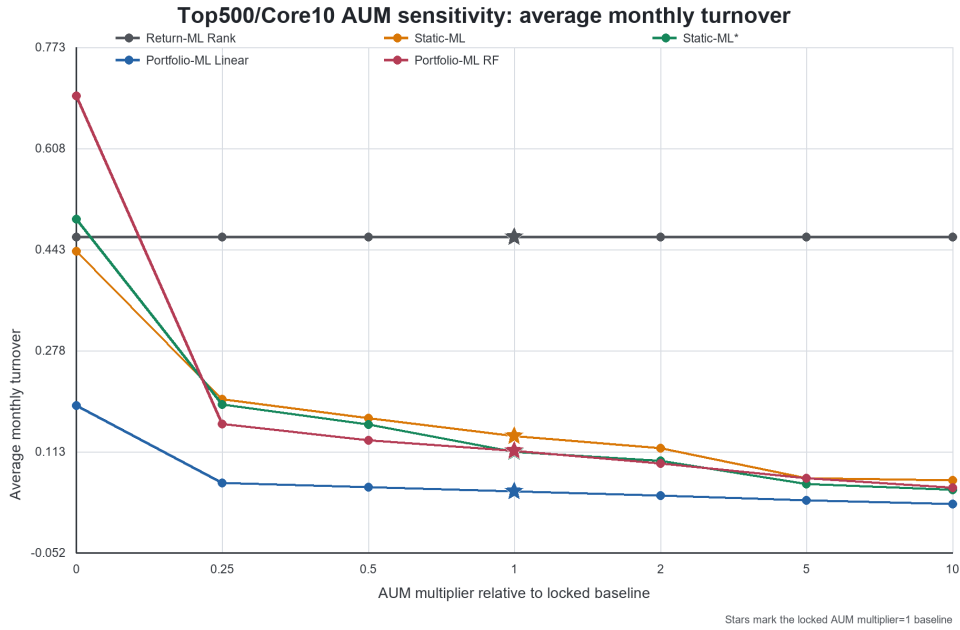


Figure B.1: Average monthly turnover across the fixed-AUM sensitivity grid. Each cost-aware point is generated by the corresponding scenario-specific cumulative portfolio experiment.

Table B.6: Yearly realised-to-predicted volatility ratios

Year	Static	Static*	PML Linear	PML RF
2005	1.20	1.10	0.76	0.82
2006	0.88	0.85	0.49	0.51
2007	1.28	1.19	0.84	0.93
2008	0.92	0.95	0.90	0.87
2009	1.04	1.03	0.75	0.95
2010	1.20	1.13	0.94	0.84
2011	1.28	1.24	0.74	0.66
2012	1.13	0.99	0.69	0.70
2013	1.87	1.46	0.92	1.02
2014	1.07	0.97	0.87	0.86
2015	1.02	0.93	0.83	0.93
2016	1.07	1.02	0.62	0.96
2017	1.08	1.03	0.74	1.00
2018	1.77	1.62	1.13	0.92
2019	0.93	0.86	0.65	0.82
2020	2.41	2.38	1.25	0.82
2021	2.69	2.19	1.27	1.60
2022	1.34	1.12	1.04	0.98
2023	1.29	1.21	1.20	1.23
2024	1.63	1.53	1.32	1.34

Table B.7: Summary of annual risk-calibration ratios

Method	Full	Median	P10	P90	Max.	Years > 2
Static-ML	1.39	1.20	0.93	1.92	2.69	2
Static-ML*	1.37	1.11	0.92	1.67	2.38	2
PML Linear	0.93	0.85	0.65	1.25	1.32	0
PML RF	0.95	0.93	0.70	1.24	1.60	0

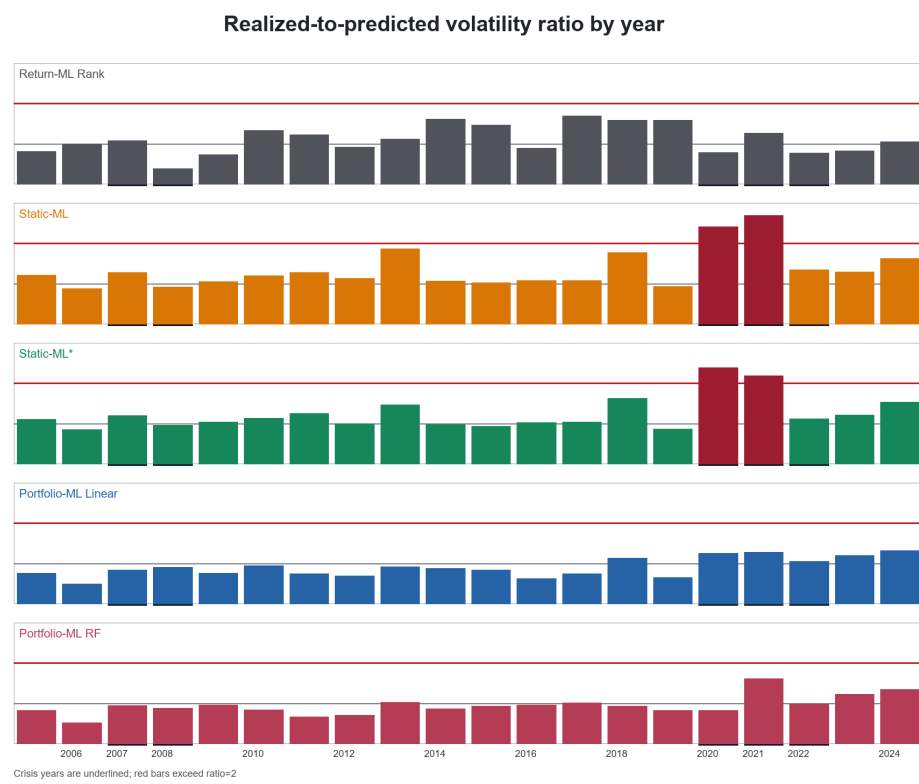


Figure B.2: Annual realised-to-predicted volatility ratios under the locked Core10 specification.

Appendix C

Supplementary Robustness and Feature Diagnostics

The final appendix collects the complete numerical evidence behind the feature-set, temporal, and signal-level robustness exercises. These diagnostics extend the baseline comparison without changing the locked Core10 result. Feature-level quantities are reported as descriptive diagnostics of the fitted policies and preserve the distinctions between raw characteristic persistence, predictive-alpha persistence, and fixed-model economic dependence.

C.1 Feature-Set Robustness

Table C.2 expands the main-text utility comparison to the full set of implementation metrics for Static-ML* and the two Portfolio-ML specifications. The positive Portfolio-ML ranking is present in Core10, Final15, and Final20 Conservative, while the magnitude and statistical precision vary across information sets. Final20 Conservative adds the ten characteristics listed in Table C.1. Its membership was fixed using 1995–2004 data-quality, pre-OOS predictive, persistence, redundancy, and theme-coverage criteria, with no use of formal 2005–2024 portfolio outcomes. The archived final artifacts do not identify an equally complete selection provenance for Final15.

Table C.1: Additional characteristics in Final20 Conservative

qmj_safety	operating_cashflow_yield
debt_to_book	operating_accruals
sales_to_market	qmj_profitability
market_beta_252d	dollar_volume_variability
momentum_18_1	ebit_to_market

Table C.3 reports the corresponding utility-difference inference. Core10 has the tightest evidence: both Portfolio-ML confidence intervals relative to Static-ML* lie above zero. The Final15 and Final20 Conservative intervals include zero, even though the point differences remain positive.

C.2 Evaluation Excluding 2007

The available 2007 sensitivity analysis is an evaluation-only exclusion conducted on Final15. It removes the twelve realised observations from 2007, leaving 228 months, and recomputes the evaluation statistics on that shortened sample. Model fits, selected hyperparameters, the adjustment matrix, and the post-2007 state path remain those of the

Table C.2: Complete performance of the principal methods across feature sets

Feature set	Method	Gross	Cost	Net	Vol.	Sharpe	Utility	Turnover	Lev.
			annualised	%	except	Sharpe		monthly	
								%	
Core10	Static-ML*	0.93	0.06	0.87	3.93	0.222	0.46	11.23	0.730
Core10	PML Linear	3.15	0.01	3.14	4.23	0.743	2.10	4.74	0.389
Core10	PML RF	3.90	0.04	3.86	5.21	0.742	2.36	11.42	0.709
Final15	Static-ML*	2.10	0.23	1.87	7.34	0.254	0.65	20.65	1.398
Final15	PML Linear	3.27	0.01	3.25	4.30	0.757	2.18	5.91	0.419
Final15	PML RF	4.07	0.04	4.03	5.43	0.742	2.53	11.59	0.697
Final20 Conservative	Static-ML*	1.84	0.08	1.76	4.72	0.372	1.22	13.32	0.881
Final20 Conservative	PML Linear	3.26	0.01	3.25	4.37	0.744	2.16	5.66	0.450
Final20 Conservative	PML RF	3.73	0.02	3.71	4.68	0.793	2.46	8.03	0.607

Table C.3: Utility-difference inference across feature sets

Feature set	Candidate	Benchmark	ΔU	HAC t	Bootstrap 95% CI	$P(\Delta > 0)$	Years won
Core10	PML Linear	Static*	1.64	2.09	[0.20, 3.06]	0.985	15/20
Core10	PML RF	Static*	1.90	2.02	[0.09, 3.63]	0.981	15/20
Core10	PML RF	PML Linear	0.26	0.42	[-1.01, 1.37]	0.670	13/20
Final15	PML Linear	Static*	1.53	0.89	[-1.06, 4.80]	0.853	14/20
Final15	PML RF	Static*	1.89	1.14	[-0.70, 4.90]	0.907	14/20
Final15	PML RF	PML Linear	0.35	0.93	[-0.31, 1.00]	0.854	13/20
Final20 Conservative	PML Linear	Static*	0.95	1.11	[-0.65, 2.53]	0.882	12/20
Final20 Conservative	PML RF	Static*	1.25	1.45	[-0.46, 2.75]	0.928	14/20
Final20 Conservative	PML RF	PML Linear	0.30	1.10	[-0.19, 0.77]	0.882	12/20

ΔU and confidence intervals are annualised utility percentage points.

original cumulative experiment. Table C.4 reports method-level outcomes before and after the exclusion, and Table C.5 reports the relative utility comparisons.

Table C.4: Final15 method-level results before and after removing realised 2007 observations

Sample	Method	Net return (% p.a.)	Volatility (% p.a.)	Utility (% p.a.)	Net Sharpe
Full 2005–2024	Static*	1.87	7.34	0.65	0.254
Full 2005–2024	PML Linear	3.25	4.30	2.18	0.757
Full 2005–2024	PML RF	4.03	5.43	2.53	0.742
Exclude 2007	Static*	3.30	5.51	2.44	0.598
Exclude 2007	PML Linear	3.49	4.27	2.45	0.817
Exclude 2007	PML RF	4.44	5.42	2.95	0.819

C.3 Economic Feature-Importance Diagnostics

The fixed-model EFI exercise neutralises one Core10 rank at 0.5 and recursively recomputes the resulting portfolio path under the fitted policy. Coefficients, selected hyperparameters, the RF mapping, risk inputs, transaction-cost inputs, γ , and AUM remain fixed; current trades, exits, costs, and inherited positions evolve along the counterfactual path. EFI is the annualised utility difference between the full-policy path and the neutralised path. Table C.6 reports the neutralised utility levels alongside the resulting EFI values. The full-policy utilities are 2.1013% for Linear and 2.3647% for Random Features.

Table C.5: Final15 utility comparisons with and without 2007 in evaluation

Sample	Comparison	ΔU	HAC t	Bootstrap 95% CI	$P(\Delta > 0)$
Full	Linear – Static*	1.53	0.89	[-1.06, 4.80]	0.853
Full	RF – Static*	1.89	1.14	[-0.70, 4.90]	0.907
Exclude 2007	Linear – Static*	0.00	0.00	[-2.16, 1.76]	0.540
Exclude 2007	RF – Static*	0.51	0.53	[-1.58, 2.28]	0.721

ΔU and confidence intervals are annualised utility percentage points.

Table C.6: Fixed-model feature neutralisation and economic feature importance

Feature	Linear neutralised U (% p.a.)	Linear EFI (pp)	RF neutralised U (% p.a.)	RF EFI (pp)
momentum_12_1	1.885	0.216	1.869	0.496
volatility_12m	2.030	0.071	2.231	0.133
idio_vol	2.031	0.070	2.396	-0.031
gross_profitability	2.075	0.026	2.277	0.088
beta_mkt	2.098	0.003	2.349	0.016
adv_6m	2.100	0.002	2.388	-0.024
log_mcap	2.104	-0.003	2.381	-0.016
reversal_1m	2.105	-0.004	2.381	-0.016
asset_growth	2.122	-0.021	2.401	-0.036
book_to_market	2.122	-0.021	2.554	-0.190

Positive EFI means that neutralising the feature lowers realised annualised utility for the fitted policy. The measure is a fixed-model economic dependence diagnostic.

C.4 Characteristic Persistence and Predictive Alpha Decay

Raw characteristic persistence is the mean monthly cross-sectional Spearman correlation between the rank at t and $t + 1$ among securities present in both months. Predictive-alpha decay uses the rank observed at t against the single-month excess return realised at horizons $h = 1, \dots, 12$. The Fama–MacBeth coefficient and Rank IC are computed separately at each horizon, the sign at $h = 1$ defines the reference direction, and signed future mass sums the sign-adjusted Fama–MacBeth coefficients over $h = 2, \dots, 12$. First half-decay is the first horizon from $h = 2$ at which the sign-adjusted coefficient falls below one half of the absolute $h = 1$ coefficient; the reversal horizon is the first negative sign-adjusted horizon, with “> 12” indicating no reversal in the window.

Table C.7: Core10 persistence, alpha-decay, and EFI summary

Feature	Raw pers.	Rank IC	FMB $h = 1$ (%)	Future mass (%)	Half-decay h	Reversal h	Linear EFI (pp)	RF EFI (pp)
adv_6m	0.996	0.0039	0.062	2.452	11	11	0.002	-0.024
idio_vol	0.996	0.0067	0.059	-0.874	4	5	0.070	-0.031
gross_profitability	0.996	0.0194	0.634	7.034	> 12	> 12	0.026	0.088
log_mcap	0.992	0.0038	-0.820	7.301	9	> 12	-0.003	-0.016
beta_mkt	0.991	0.0056	0.399	2.609	7	> 12	0.003	0.016
volatility_12m	0.988	0.0010	0.289	2.764	9	> 12	0.071	0.133
book_to_market	0.987	0.0148	-0.444	5.497	> 12	> 12	-0.021	-0.190
asset_growth	0.933	0.0028	0.146	-2.040	2	4	-0.021	-0.036
momentum_12_1	0.891	0.0161	0.696	3.369	6	> 12	0.216	0.496
reversal_1m	-0.010	0.0072	0.062	2.297	2	7	-0.004	-0.016

The complete horizon profiles are shown in Figure C.1. Momentum is relatively fast-moving in raw characteristic space, with persistence rank 9 of 10, yet its $h = 1$ Fama–MacBeth slope is 0.696 percentage points, its signed future mass is 3.369 percentage points,

its first half-decay occurs at $h = 6$, and no Fama–MacBeth reversal appears through $h = 12$. One-month reversal has near-zero raw persistence, first reaches the half-decay threshold at $h = 2$, and first changes Fama–MacBeth direction at $h = 7$. These profiles illustrate the distinction between state persistence, predictive persistence, and portfolio-level economic dependence.

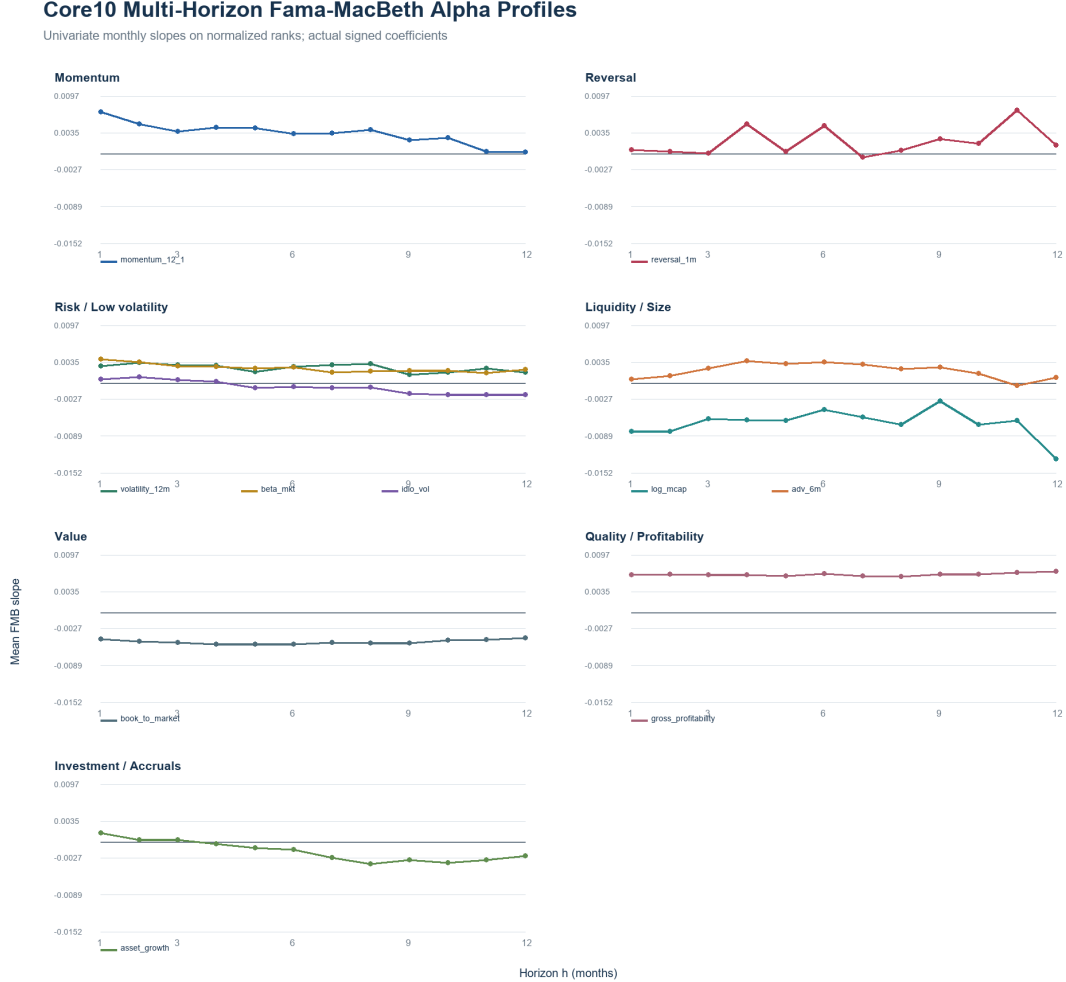


Figure C.1: Multi-horizon Fama–MacBeth predictive-alpha profiles for the ten Core10 characteristics. Each path uses the characteristic rank observed at t and the incremental one-month excess return at future horizon h .

C.5 Prespecified Persistence Diagnostics

The persistence extension used a prespecified four-part diagnostic gate to assess whether predictive-alpha persistence provided a materially stronger and stable explanation of EFI than raw characteristic persistence. Table C.8 records the criteria and outcomes. None of the four conditions is satisfied, giving the final classification *mixed*. This classification applies to the descriptive feature-level exercise and does not constitute a structural test of the Gârleanu–Pedersen model.

Across the ten features, the raw-persistence Spearman correlation with EFI is 0.212 for Linear and -0.091 for Random Features. The primary Fama–MacBeth signed-future-mass correlation is 0.115 for Linear and 0.430 for Random Features, with materially different leave-one-feature-out stability across the two models. The evidence therefore supports the

Table C.8: Prespecified persistence diagnostic gate

Criterion	Assessment	Result
Material improvement	The Linear FMB signed-future-mass correlation was required to exceed the raw-persistence correlation by at least 0.20. The observed values are 0.115 and 0.212, respectively.	Fail
Cross-metric/model consistency	The primary relation was required to be positive in both Portfolio-ML models, with at least 60% of the prespecified metric signs positive within each model. The observed sign pattern does not satisfy this condition.	Fail
Leave-one-out stability	The FMB signed-future-mass relation was required to retain its sign in both models with a maximum absolute leave-one-out change no greater than 0.35. The Linear relation changes sign.	Fail
Momentum/reversal coherence	Both signals were required to satisfy the prespecified persistence and decay conditions. Momentum satisfies its subcondition; reversal subsequently recovers and its $h = 1$ Fama-MacBeth and Rank-IC signs disagree.	Fail

narrower conclusion used in the main text: signal persistence is economically relevant as a diagnostic, but it does not provide a stable one-dimensional explanation of Portfolio-ML feature dependence in this sample.

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